



Northumbria
University
NEWCASTLE

**Hybrid Temporal Deep Learning for Three-Class Alzheimer's
Disease Staging: A Controlled Comparison of BiLSTM-Transformer
and TCN-Transformer Architectures on Longitudinal ADNI Data**

Student Name: Adam Buick

Student ID: w21002232

Supervisor Name: Sharnil Pandya

BSc Computer Science with Artificial Intelligence

KNV6013 Computing Project

Date: 01/05/2026

Word Count: 13195

Acknowledgements

I would like to thank my supervisor, Dr Sharnil Pandya, for his guidance, feedback, and continued support throughout this project. I am grateful to the Alzheimer's Disease Neuroimaging Initiative (ADNI) for providing the longitudinal dataset that made this research possible. Finally, I would like to thank my family and friends for their encouragement and support throughout my final year, and my entire period of studying at the university.

Abstract

Alzheimer's Disease affects approximately 55 million people globally and represents one of the most significant challenges in modern healthcare, with prevalence projected to nearly triple by 2050. Early and accurate classification of disease progression remains clinically critical, yet existing computational approaches are predominantly cross-sectional, capturing a single point in time rather than the progressive neuroanatomical trajectory that defines the disease. This limitation is most pronounced at the Mild Cognitive Impairment stage, where the risk of conversion to dementia is highest but static models consistently fail to discriminate between stable and progressive cases.

This study developed and evaluated three deep learning architectures for three-class CN/MCI/Dementia staging using longitudinal MRI and clinical data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), comprising 580 subjects across 1,711 visits. A Bidirectional Long Short-Term Memory (BiLSTM)-Transformer and a Temporal Convolutional Network (TCN)-Transformer were implemented as the primary hybrid temporal models, with a Simple ANN serving as a cross-sectional deep learning baseline. All three architectures shared an identical 3D CNN encoder for MRI feature extraction and were trained under identical conditions using Focal Loss, the Adam optimiser, and early stopping, ensuring that performance differences were attributable to temporal modelling mechanism rather than confounding variation. Multiple Imputation by Chained Equations (MICE) was applied to address the 44.7% subject-level missingness characteristic of longitudinal clinical cohorts, and Local Interpretable Model-Agnostic Explanations (LIME) explainability analysis was applied post-training to identify the clinical features driving predictions across all three models.

The BiLSTM-Transformer achieved the highest test accuracy at 88.51%, outperforming the TCN-Transformer at 87.36% and the Simple ANN at 79.31%. Neither temporal model dominated unconditionally with the BiLSTM leading on accuracy, F1, and CN recall, while the TCN led on AUC, Dementia recall, and test loss, reflecting complementary architectural strengths rooted in their respective temporal encoding mechanisms. Both temporal models substantially outperformed the Simple ANN baseline, with the most clinically significant advantage at the MCI stage where the ANN's 66.67% recall improved to 80.95% for the BiLSTM-Transformer. MICE imputation provided meaningful accuracy gains for both temporal models, while the cross-sectional ANN showed no benefit. LIME analysis identified CDRSB as the dominant feature across all three architectures, with consistent placement of established cognitive and neuroimaging markers within the top five features confirming that models attended to clinically meaningful signals.

The study provides the first direct controlled comparison of BiLSTM-Transformer and TCN-Transformer hybrid architectures on longitudinal ADNI data within a single experiment, alongside evidence that longitudinal sequence modelling adds value over static deep learning for three-class AD staging. Calibration limitations were identified across all models, with post-hoc temperature scaling recommended before clinical translation.

Contents

Title Page	1
Acknowledgements	2
Abstract.....	3
Introduction.....	6
Research and Planning.....	7
Existing Literature	7
Alzheimer's Disease and the Clinical Problem.....	7
Traditional and Early ML Approaches to AD Prediction	7
Longitudinal and Multimodal Approaches	8
Recurrent and Temporal Architectures in Healthcare	9
Transformer Architectures and Hybrid Models.....	11
Explainability in Clinical AI.....	12
Summary and Gap Analysis.....	13
Discussion of Approaches	13
Tools and Techniques	14
Practical Work	15
Data Collection Process.....	15
Data Analysis and Preprocessing	15
MICE Imputation	15
Feature Engineering	16
Train/Validation/Test Split.....	17
Model Architecture	18
LIME Explainability Analysis.....	27
Ethical Considerations	27
Results.....	28
Simple ANN Metrics	28
BiLSTM-Transformer Metrics	31
TCN-Transformer Metrics	35
Comparison Metrics.....	38
Discussion and Evaluation of Results	48
Interpretation of Results.....	48
LIME and Clinical Relevance.....	49
Calibration and Clinical Deployment	50
Confidence in Findings and Generalisability	51
Limitations of the Study.....	51

Evaluation of the Project Process.....	52
Legal, ethical, professional.....	52
Conclusion and Recommendations.....	53
Research Summary	53
Contribution.....	53
Recommendations	53
References	54
Appendix	59

Introduction

Alzheimer's Disease represents one of the most pressing challenges in modern healthcare, affecting an estimated 55 million people globally and projected to reach 139 million by 2050 (WHO, 2025; GBD 2019 Dementia Forecasting Collaborators, 2022). Despite years of research, early and accurate classification remains difficult, particularly at the Mild Cognitive Impairment stage where the risk of progression to dementia is highest but clinical indicators are least discriminative. Traditional machine learning approaches have demonstrated strong performance in binary classification tasks but consistently fail at the MCI stage due to their cross-sectional nature, capturing a single point in time rather than the progressive neuroanatomical trajectory that defines the disease (Bhagwat et al., 2018; Cuingnet et al., 2011).

This study addressed this limitation by developing and evaluating three deep learning architectures for three-class CN/MCI/Dementia staging using longitudinal MRI and clinical data from the ADNI. A BiLSTM-Transformer and a TCN-Transformer hybrid architecture were implemented as the primary models, with a Simple ANN serving as a cross-sectional baseline. All three models were trained under identical experimental conditions using late fusion of 3D MRI embeddings and tabular clinical features, with MICE applied to address the substantial missing data characteristic of longitudinal cohorts. LIME explainability analysis was applied post-training to identify the clinical features driving predictions across all three architectures.

The BiLSTM-Transformer achieved the highest test accuracy at 88.51%, followed by the TCN-Transformer at 87.36% and the Simple ANN at 79.31%, with temporal models demonstrating the largest advantage at the MCI stage. The study addressed this limitation by developing a controlled comparison of three architectures under identical conditions, with findings presented across the Research and Planning, Practical Work, Results, and Discussion sections that follow.

Research and Planning

Existing Literature

Alzheimer's Disease and the Clinical Problem

Alzheimer's Disease represents an important global health crisis, currently serving as the most common cause of cognitive impairment and dementia in individuals over 65 (Atri, 2019; Kumar et al., 2022). Approximately 55 million people are currently living with dementia globally, with Alzheimer's disease accounting for 60–70% of cases (WHO, 2021; Cummings, Morstorf and Zhong, 2014), and this figure is projected to reach 139 million by 2050 (WHO, 2025; GBD 2019 Dementia Forecasting Collaborators, 2022). The associated economic burden is projected to reach US \$2 trillion by 2030, falling predominantly on informal caregivers (Tay et al., 2024).

The clinical trajectory is divided into three stages: Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Dementia (Kumar et al., 2022). CN individuals perform within expected range on cognitive tests and retain independent function. MCI is characterised by cognitive performance below expected range with mild functional impact on complex daily activities, while Dementia involves clearly evident functional impairment such that the individual is no longer fully independent (Jack et al., 2018).

Early and accurate detection is critical, as timely identification preserves the opportunity for intervention while patients can still function independently, despite no disease-modifying treatments currently being available (Rasmussen and Langerman, 2019). MCI represents a particularly challenging diagnostic target, as not all individuals progress to dementia, and the lack of universally adopted diagnostic criteria means prevalence estimates across studies range from 3% to 42% (Jack et al., 2018; Anderson, 2019). These factors make reliable MCI classification both clinically significant and technically difficult, directly motivating the computational approach taken in this dissertation.

Traditional and Early ML Approaches to AD Prediction

Before the application of machine learning to AD classification, clinical diagnosis followed a multi-stage process beginning with cognitive screening tools such as the Mini-Mental State Examination (MMSE) and the Montreal Cognitive Assessment (MoCA), progressing through structural MRI and ultimately biomarker confirmation via amyloid PET or CSF analysis of A β 42 and tau proteins (Porsteinsson et al., 2021; Anderson, 2019). This process is inherently challenging as the subtle and variable emergence of symptoms makes early recognition difficult, particularly in primary care settings where clinicians have limited time and where both patients and providers frequently dismiss early cognitive changes as part of the normal ageing process (Porsteinsson et al., 2021). Critically, a definitive diagnosis of AD remains impossible during life, as only post-mortem examination can confirm the disease through detection of amyloid- β plaque deposition and tau protein, with diagnosis further complicated by co-pathologies including cerebrovascular disease, Lewy body pathology, and TDP-43 deposits present in the majority of confirmed cases (Porsteinsson et al., 2021; DeTure and Dickson, 2019). Furthermore, traditional clinical assessment is fundamentally cross-sectional, capturing a single point in time rather than modelling the progressive trajectory of the disease, a critical limitation given that AD pathology develops over years or decades before symptoms emerge (Sperling et al., 2011; Bhagwat et al., 2018).

Early machine learning research applied to AD classification was dominated by Support Vector Machines (SVMs), Random Forests (RFs), and simple Artificial Neural Networks (ANNs), predominantly trained on cross-sectional neuroimaging and clinical data from ADNI (Sarica, Cerasa and Quattrone, 2017; Alatrany et al., 2024; Lazli, 2025; Bhagwat et al., 2018). SVMs became one of the earliest and most widely adopted approaches, with whole-brain methods achieving up to 81% sensitivity and 95% specificity in distinguishing AD patients from healthy controls and demonstrating strong generalisation across data types and cohorts (Cuingnet et al., 2011; Alatrany et al., 2024). Recent comparative studies continue to utilise ANNs and SVMs combined with dimensionality reduction techniques such as PCA, achieving satisfactory performance in binary classification tasks (Lazli, 2025). Random Forests offer the additional advantage of intrinsic feature importance ranking, with research consistently identifying the hippocampus, amygdala, and inferior lateral ventricle as the most discriminative biomarkers for AD classification (Song et al., 2021; Gray et al., 2013; Cuingnet et al., 2011). Utilising regional volumes from structural MRI alone, these ensemble models achieved accuracies of up to 90.2% in large ADNI cohorts, demonstrating performance comparable to deep learning models while maintaining greater stability with fewer features (Song et al., 2021; Gray et al., 2013).

Despite these high accuracies in binary diagnostic tasks, these traditional ML approaches face a significant prognostic ceiling when applied to the intermediate MCI stage. Because these models are fundamentally cross-sectional, they rely on a single snapshot of data that fails to account for the nonlinear longitudinal symptom progression inherent in the AD continuum (Bhagwat et al., 2018; Sperling et al., 2011). Research evaluating ten different high-dimensional methods on the ADNI database found that while whole-brain SVMs excel at identifying established AD, no classifier obtained results significantly better than chance when predicting MCI-to-AD conversion (Cuingnet et al., 2011). Static models using baseline attributes often merely predict a continuation of current symptom severity rather than future pathological trajectory, with instability particularly pronounced for patients in the mid-range of clinical performance whose baseline scores offer little predictive power regarding whether they will remain stable or decline (Bhagwat et al., 2018). This is directly evidenced by Bhagwat et al. (2018), whose cross-sectional models consistently failed to identify the slow-declining trajectory class, representing patients whose MCI progresses to Alzheimer's disease, with ANN-based approaches failing to predict this class entirely in multiple cross-validation folds. Consequently, the inability of cross-sectional baselines to capture subtle neuroanatomical changes indicative of long-term decline directly motivates the transition toward longitudinal and multimodal modelling approaches (Bhagwat et al., 2018).

Longitudinal and Multimodal Approaches

The transition from traditional machine learning to longitudinal modelling is necessitated by the inherent nature of Alzheimer's Disease (AD), where pathological changes begin before the onset of noticeable clinical symptoms (Atri, 2019; Sperling et al., 2011; Jack et al., 2018). Computational models that rely on cross-sectional "snapshots" are fundamentally limited as they cannot account for nonlinear longitudinal symptom progression that defines the AD continuum (Bhagwat et al., 2018; Jack et al., 2018; Lei et al., 2020). Longitudinal modelling is critical because it allows the model to capture the rate of neuroanatomical decline and learn change patterns across time (Bhagwat et al., 2018; Ding et al., 2023). Research (Ding et al., 2023) has directly quantified this longitudinal advantage, with a Long Short-Term Memory (LSTM) model achieving an AUC of 0.93 for MCI-to-AD conversion prediction, significantly outperforming cross-sectional Random Forest baselines.

The case for multimodal data integration is equally well-established. No single biomarker captures the full complexity of AD pathology. Structural MRI reveals neuroanatomical atrophy, cognitive assessments such as the MMSE and CDRSB track functional decline, and genetic markers such as APOE4 provide risk stratification information that imaging alone cannot supply (Venugopalan et al., 2021). A large-scale multimodal deep learning study by Qiu et al (2022) demonstrated that fusion models achieved classification accuracy comparable to practising neurologists, a benchmark that unimodal models consistently failed to reach. Furthermore, a three-year prospective study confirmed that plasma p-tau217, CSF, and amyloid PET each contributed independent and complementary information, with no single modality capturing the full picture of subtle cognitive decline (López-Martos et al., 2025). Multimodal approaches have been shown to raise MCI conversion prediction accuracy from 75% with single-modality data to 81% when longitudinal multimodal inputs are incorporated (Lee et al., 2019). Collectively, these findings indicate that the diagnostic limitations of any single biomarker domain are not just technical but reflect the genuinely multidimensional nature of AD pathology.

A practical challenge is the pervasive presence of missing data. Longitudinal cohort studies such as ADNI are characterised by participant dropout, irregular visit schedules, and incomplete modality acquisition (Aghili et al., 2022). Critically, this missingness is not random, with poor cognitive performance at baseline predicting loss of follow-up, meaning patients who drop out tend to have more pronounced AD-like profiles at withdrawal (Lo and Jagust, 2012). Simply excluding incomplete cases systematically removes the most clinically informative subjects, introducing selection bias that undermines the validity of any downstream model (Austin et al., 2021). Research evaluating imputation strategies specifically on ADNI multimodal data has demonstrated that MICE produces superior classification performance compared to simpler strategies such as mean and median substitution (Aracri et al., 2025).

Taken together, the literature consistently supports two converging conclusions: that temporal modelling outperforms cross-sectional classification for AD staging, and that multimodal feature integration yields superior discriminative power over any single biomarker domain. Both advantages rely on robust missing data handling as a prerequisite for reliable longitudinal analysis. The question of which specific temporal architectures are best suited to exploit this structure is addressed in the next section.

Recurrent and Temporal Architectures in Healthcare

The limitations of cross-sectional approaches established in the previous section motivated the adoption of recurrent architectures capable of modelling sequential patient data across time. LSTM networks, introduced by Hochreiter and Schmidhuber (1997), represented a foundational advancement in this regard by addressing the vanishing gradient problem that rendered earlier Recurrent Neural Networks (RNNs) ineffective for capturing long-range dependencies in sequential data. The original architecture incorporated input and output gates to control information flow, with Gers, Schmidhuber and Cummins (1999) subsequently introducing the forget gate. This enabled LSTM cells to adaptively reset their internal state, completing the three-gate architecture now considered standard. Together, these gating mechanisms allow LSTMs to selectively retain and discard information across time steps, making them architecturally well-suited to longitudinal clinical datasets where disease-relevant signals accumulate gradually across multiple visits rather than being fully apparent at any single timepoint.

Application of LSTMs to AD prediction on longitudinal ADNI data demonstrated clear advantages over contemporary cross-sectional baselines. Hong et al. (2019) evaluated a longitudinal LSTM

model against three competing approaches; an SVM with sparse regression features, a Random Forest combined with a Deep Neural Network, and a CNN with PCA-LASSO feature selection, using MRI time-series data from 1,105 ADNI subjects. The LSTM achieved an AUC of 0.935 for AD versus NC and an mAUC of 0.777 for three-class prediction, outperforming all cross-sectional methods on these tasks. Critically, however, the LSTM performed worst on MCI versus NC classification, achieving an AUC of only 0.697 which is below all three baselines on that task. The authors explicitly attributed this to the model's inability to distinguish stable MCI from progressive MCI, as these two subgroups are clinically similar at baseline and only diverge over time. This finding is significant because it identifies an inherent limitation of unidirectional LSTM architectures when applied to the most diagnostically challenging stage of the disease and directly motivates the adoption of more powerful temporal architectures capable of exploiting bidirectional and multi-scale temporal dependencies.

Bidirectional LSTMs (BiLSTMs) address the directional limitation of unidirectional LSTMs by processing sequences in both forward and backward directions simultaneously, allowing the model to incorporate both past and future context when encoding each time step (Schuster and Paliwal, 1997). In the context of longitudinal clinical data, this is particularly valuable as later visits can contextualise the significance of earlier measurements and patterns of decline may only become interpretable when the trajectory across all timepoints is considered together. Jomeiri et al. (2024) evaluated a hybrid DenseNet-BiLSTM framework on 684 longitudinal ADNI subjects scanned over 48 months across seven timepoints. On the clinically critical sMCI versus pMCI task, where sMCI denotes stable MCI patients who do not convert to Alzheimer's Disease and pMCI denotes progressive MCI patients who eventually do, the standalone BiLSTM demonstrated superior diagnostic reliability, achieving an accuracy of 79.45%, which significantly outperformed the VGG19 (72.12%) and DenseNet (76.94%) spatial baselines. While VGG19 and ResNet50 recorded higher AUC values in this specific experiment (88.02 and 89.20 respectively), their lower accuracy and sensitivity suggest a reduced ability to correctly identify true progressors compared to architectures that incorporate temporal dependencies. Ultimately, the study concluded that the synergistic integration of spatial and temporal features allowed the Hybrid DenseNet-BiLSTM to reach the highest classification accuracy of 83.51%, identifying it as the most effective framework for Alzheimer's disease staging.

TCNs offer a fundamentally different approach to sequential data modelling, replacing the recurrent connections of LSTMs with dilated causal convolutions that allow the receptive field to expand exponentially with network depth (Bai, Kolter and Koltun, 2018). Unlike recurrent architectures, which process sequences step-by-step and are susceptible to vanishing gradients over long time horizons, TCNs apply convolutional filters in parallel across the entire input sequence, preserving strict temporal causality by ensuring that predictions at any time step depend only on past observations. These architectural properties amount to several practical advantages over recurrent models. TCNs require significantly less memory for long input sequences, as convolutional kernels are shared across a layer rather than storing partial results across multiple gates; they allow greater parallelisation since the entire sequence is processed simultaneously rather than step-by-step; and they are generally easier to train, avoiding the vanishing gradient instability commonly encountered in LSTM-based architectures (Thapaliya et al., 2024). Beyond theory, these advantages have been validated in direct comparisons with recurrent architectures. Gopali et al. (2021) evaluated TCN and LSTM models of equivalent depth on multivariate time-series anomaly detection, finding that the TCN achieved a higher F1 score than both single and three-layer LSTM variants (0.920 vs. 0.875 and 0.901, respectively), while showing smoother validation loss convergence and reaching a stable model in fewer training

epochs. Within neuroimaging specifically, Thapaliya et al. (2024) incorporated a TCN block into a graph transformer architecture for Alzheimer's disease classification using rsfMRI data from ADNI, demonstrating that TCNs can effectively extract temporal features from longitudinal brain activity data.

Given these properties, TCNs represent a compelling architecture for neurodegenerative disease modelling, where the progressive and temporally structured nature of conditions such as Alzheimer's disease demands robust capture of long-range clinical trajectories across extended follow-up periods.

Transformer Architectures and Hybrid Models

While recurrent and convolutional architectures capture local and sequential temporal patterns effectively, neither architecture explicitly models the relative importance of different timepoints within a sequence. The Transformer architecture, introduced by Vaswani et al. (2017), addresses this through a self-attention mechanism that calculates relationships between all positions in a sequence simultaneously. A defining trait of the architecture is that each input token flows through its own computational path while remaining directly dependent on every other token in the sequence (Amatriain et al., 2023). This allows the model to draw global dependencies regardless of temporal distance, offering higher connectivity and lower computational complexity than traditional sequential models (Vaswani et al., 2017; Amatriain et al., 2023). In the context of longitudinal clinical data, this is particularly valuable as patient visits do not carry equal diagnostic significance. For example, a visit capturing an abrupt increase in hippocampal atrophy or a rapid decline in cognitive scores may be far more informative than stable intermediate intervals (Chen et al., 2023). Experimental evidence supports these theoretical advantages in the AD domain specifically. Moghaddami, Schubring and Siadat (2025) directly compared a Transformer model against LSTM, GRU, and minRNN baselines on longitudinal ADNI visit sequences for three-class CN/MCI/AD prediction, finding that the Transformer achieved a higher mAUC of 0.920 and F1 score of 0.824 than all recurrent baselines. Critically, the Transformer demonstrated a much greater ability to identify subjects who converted from MCI to AD, achieving 64.8% accuracy on converter sequences compared to just 21.7% for LSTM, confirming that attention-based global sequence modelling is particularly well-suited to detecting the subtle longitudinal patterns that characterise disease progression.

Pure Transformer architectures are however data-hungry by design. They impose no prior assumptions about how features relate to one another, meaning the burden of learning every relationship falls entirely on the training data (Chen et al., 2023). This is a requirement that has tension with the reality of clinical research, where longitudinal datasets are costly to collect and subject numbers are necessarily modest. Chen et al (2023) demonstrated this directly through a study using ADNI, finding that a pure 3D Vision Transformer achieved only 80.24% accuracy on AD classification. This is more than thirteen percent below their hybrid CNN-Transformer architecture which achieved 93.43% on the same dataset. This performance gap illustrates that while attention mechanisms are theoretically powerful, standalone transformers struggle to extract meaningful representations from limited clinical data without the prior knowledge that convolutional or recurrent encoders provide. Hybrid architectures address this by dividing the modelling task between specialist components. A temporal encoder handles sequential feature extraction efficiently on limited data, while a transformer attention mechanism independently models global dependencies, with both representations combined for the final classification decision (Bahroun and Jeon, 2026). This design philosophy has been tested in terms of AD, with Thapaliya et al. (2024) incorporating a TCN block within a graph transformer architecture for AD

classification on ADNI data. Their model achieved an AUC of 0.76 and an F1 score of 0.68, significantly outperforming benchmarks such as BrainRGIN (AUC 0.64) and traditional Graph Transformers (AUC 0.68), demonstrating that convolutional temporal feature extraction and attention-based global modelling are complementary rather than competing approaches. It is this hybrid design rationale that motivates the architectural approach taken in the present study, combining temporal encoders with transformer attention mechanisms to exploit longitudinal clinical structure while remaining suitable for the limited dataset sizes typically found in longitudinal clinical research.

Explainability in Clinical AI

The adoption of deep learning in clinical decision support has been constrained not only by technical performance but by the opacity of the models involved. Vimbi, Shaffi and Mahmud (2024, p. 2) observe that AI models are regarded by medical practitioners as black boxes due to their inability to provide justifiable reasons for their predictions, creating a fundamental barrier to deployment. Amann et al. (2022, p. 3) extend this point, arguing that explainability enables clinicians to judge whether a system's output is trustworthy and verify that predictions are not driven by artefacts in the training data. Rosenbacke et al. (2024, p. 5) further note that in clinical settings, explanations are most effective at building trust when they are clear, concise, and directly relevant to practice. These concerns are particularly important in neurodegenerative disease such as AD, where classification decisions carry significant prognostic weight and where clinician confidence in model outputs is essential for any meaningful translation to practice.

Within XAI (Explainable Artificial Intelligence), a key distinction exists between global methods, which characterise model behaviour across the entire input distribution, and local methods, which explain individual predictions by examining model behaviour in the vicinity of a specific instance (Vimbi, Shaffi and Mahmud, 2024, p. 7). For clinical progression classification, local explanations are particularly meaningful as they allow interrogation of why a specific patient was assigned a particular stage, rather than relying on population-level tendencies. A further distinction separates model-specific methods such as Layer-wise Relevance Propagation (LRP), which requires access to internal gradients from model-agnostic approaches that can be applied to any trained classifier without modification (Vimbi, Shaffi and Mahmud, 2024., 2024, pp. 7–8). SHAP is a prominent model-agnostic alternative that assigns Shapley values to features based on game-theoretic principles and is widely used in AD research, however, its global orientation makes it less suited to instance-level clinical decisions compared to methods explicitly optimised for local instance-level approximation (Vimbi, Shaffi and Mahmud, 2024, p. 9).

LIME serves as a primary example of a local, model-agnostic explanation tool. Ribeiro et al. (2016) describe LIME as generating local explanations by perturbing inputs around a target instance and fitting an interpretable approximation to identify influential features. Its model-agnostic nature is particularly advantageous when evaluating diverse neural network architectures, as it allows a direct, standardised comparison of feature importance across architecturally distinct classifiers by treating each original model as a "black box" (Ribeiro et al., 2016). This local scope aligns with the nature of clinical staging decisions, where a justification for an individual patient's diagnosis is required to enable clinician trust and support algorithm-based decisions (Vimbi, Shaffi and Mahmud, 2024). However, because LIME relies on post-hoc approximations, Amann et al. (2022) and Ribeiro et al. (2016) warn that such explanations may not always be fully faithful to complex, non-linear decision boundaries. Consequently, while these interpretations provide critical insights into model behaviour, they should be interpreted as indicative rather than definitive in a clinical context.

Summary and Gap Analysis

The literature reviewed in the preceding sections supports several well-evidenced conclusions. Cross-sectional approaches, while effective for binary classification fail to capture the nonlinear trajectories that define disease progression, whereas longitudinal temporal architectures consistently outperform static baselines. Multimodal integration consistently demonstrates superior discriminative power over single-modality approaches, and robust missing data handling such as MICE is a prerequisite for reliable longitudinal analysis (Aracri et al., 2025). Finally, at the output stage, explainability methods such as SHAP or LIME are essential to bridge the "trust gap" between clinicians and "black-box" systems by translating model weights into biologically meaningful justifications (Fabietti et al., 2023).

Despite this breadth of evidence, a critical methodological gap persists. No study has directly compared BiLSTM-Transformer and TCN-Transformer architectures within a single controlled experiment on longitudinal ADNI data, and many existing studies applying LIME to AD classification have done so for individual models rather than comparatively across multiple architectures. The present study addresses both gaps by evaluating all three architectures under identical experimental conditions and applying LIME consistently across each model, enabling both performance benchmarking and cross-architecture feature interpretation on the same dataset.

Discussion of Approaches

The literature reviewed in the previous section established that longitudinal multimodal models consistently outperform single modality models when applied to the prediction of AD. The selection of the BiLSTM-Transformer and TCN-Transformer primary models, and the baseline ANN model, follows directly from this evidence.

Both primary architectures adopt a hybrid design philosophy motivated by the findings of Chen et al. (2023), who demonstrated that a pure Vision Transformer achieved 80.24% accuracy on ADNI compared to 93.43% for a hybrid CNN-Transformer, confirming that standalone attention mechanisms struggle on clinical scale datasets without the inductive biases provided by convolutional or recurrent encoders. By pairing a temporal encoder with a Transformer attention mechanism, each architecture is designed to exploit longitudinal structure efficiently while remaining suited to the dataset sizes typical of clinical research. The BiLSTM was selected over a unidirectional LSTM specifically because bidirectional processing allows later visits to contextualise earlier measurements, this has real value in retrospective longitudinal classification where the full sequence of visits is available at prediction time (Jomeiri et al., 2024). The TCN was selected as the contrasting temporal architecture because its dilated causal convolutions process sequences in parallel across multiple temporal scales simultaneously, processing the data in a fundamentally different way to the recurrent approach and enabling a direct controlled comparison between sequential memory-based and parallel convolutional temporal modelling on the same dataset (Bai, Kolter and Koltun, 2018; Gopali et al., 2021).

An ANN was selected as the baseline model for this comparison rather than a traditional machine learning model such as an SVM or Random Forest, as the intention was to investigate the contribution of temporal modelling specifically rather than compare deep learning against classical methods. By using a non-temporal deep learning model against two hybrid temporal architectures, the study operates on two simultaneous comparisons. The first looks at whether temporal modelling adds value over static deep learning, and the second determines which form

of temporal modelling performs better, allowing both questions to be answered within a single controlled experiment. This decision is supported by the established use of feedforward networks as deep learning baselines in AD classification literature, where they provide a performance lower bound against which the temporal models can be directly compared (Lazli, 2025; Bhagwat et al., 2018).

While OASIS-3 was considered as an alternative, ADNI was selected as the primary dataset due to its superior longitudinal standardisation. ADNI's standardised acquisition protocol, providing paired longitudinal MRI and clinical assessments across multiple visits per subject, directly supports the multimodal longitudinal approach taken in this study. A concern with OASIS-3 is the significant variability in scanning protocols across different acquisition sites, which can introduce technical inconsistencies that complicate model training (Bahroun and Jeon, 2026). By focusing on ADNI's standardised framework, this study ensures a controlled environment for architectural comparison, while external validation on OASIS-3 remains a priority for future work to assess model generalisability across diverse clinical populations (Bhagwat et al., 2018; Jomeiri et al., 2024).

Tools and Techniques

Python was selected as the primary development language given its dominant adoption in deep learning research. PyTorch was used to implement and train all three architectures, providing dynamic computation graphs, native Focal Loss support, and sequence handling utilities including `pack_padded_sequence` for the BiLSTM, attention masking for the Transformer encoder, and mask mean pooling for the TCN. Scikit-Learn provided the `IterativeImputer` for MICE imputation using a `RandomForestRegressor` estimator, alongside `StandardScaler` for feature normalisation and `train_test_split` for dataset partitioning. The LIME library's `LimeTabularExplainer` was selected for its local, model-agnostic design, enabling standardised feature importance analysis across all three architectures. NiBabel loaded NIfTI MRI files from the ADNI file structure, with SciPy providing volumetric resizing to 64×64×64. Matplotlib generated all training curves, confusion matrices, and performance visualisations. NumPy provided the numerical foundation for array processing and LIME score aggregation. A random seed of 42 was used across all models for reproducibility. Development was carried out in Visual Studio Code with Git source control.

Practical Work

Data Collection Process

The data used in this study was provided by the ADNI database (adni.loni.usc.edu). Access required a formal application describing the intended research purpose, followed by agreement to the ADNI Data Use Agreement (DUA). Approval was granted one month after submission, after which the relevant longitudinal clinical and neuroimaging data were downloaded directly from the repository.

Examination of the dataset revealed 580 subjects across three diagnostic classes: cognitively normal (CN, 31.2%), mild cognitive impairment (MCI, 24.3%), and dementia (44.5%), summarised in Figure 1. Longitudinal coverage averaged 3.0 visits per subject over a mean follow-up period of 11.7 months, producing 1,711 total visits with visit counts ranging from two to four. Within the subjects, 75.7% maintained a stable diagnosis across visits, while 22.4% exhibited clinical decline, reflecting realistic disease progression patterns within the ADNI cohort.

Data Analysis and Preprocessing

Nine clinical features were utilised, consisting of continuous cognitive assessments (MMSE, CDRSB), MRI-derived volumetric measures (Hippocampus, Ventricles, WholeBrain, ICV), a continuous demographic variable (Age), a discrete education variable (PTEDUCAT), and a binary genetic risk indicator (APOE4). Exploratory analysis revealed substantial amounts of missing data concentrated in MRI-derived features, most notably Hippocampus volume (23.9% of visits), Ventricles (6.2%), WholeBrain (4.8%), and ICV (3.2%), meaning only 55.3% of subjects presented with fully complete longitudinal records.

Class Distribution — ADNI Dataset

Diagnosis Class	Count
CN (Cognitively Normal)	181 (31.2%)
MCI (Mild Cognitive Impairment)	141 (24.3%)
Dementia	258 (44.5%)
Total Subjects	580

Figure 1: Class distribution across the ADNI dataset.

MICE Imputation

As identified in the preceding section, MRI-derived missingness across 44.7% of subjects necessitated a principled imputation strategy prior to model training. Two conventional remedies were considered and rejected. Listwise deletion would have eliminated 44.7% of the dataset, severely limiting training data, whilst mean imputation ignores the inter-feature correlations present in neuroimaging data, introducing systematic bias.

MICE was selected as it models each incomplete feature as a function of all remaining features iteratively, making it well-suited to a multivariate dataset where MRI-derived measures are known

to correlate with cognitive assessments such as MMSE and CDRSB (Van Buuren, 2018). A Random Forest estimator was used within the MICE framework over 10 iterations, providing robustness to non-linear relationships between features. To maximize the utility of the available longitudinal signals and ensure the regression weights for imputation were informed by the largest possible distribution, the imputer was fitted to the entire dataset prior to partitioning. As shown in Figure 2, post-imputation feature completeness reached 100% across all 580 subjects and 1,711 visits, providing a robust and continuous data sequence for subsequent temporal modelling.

Missing Data: Pre-MICE vs Post-MICE Imputation

Feature	Missing Per Visit — Pre-MICE	Missing Per Visit — Post-MICE
AGE	0 (0.0%)	0 (0.0%)
MMSE	0 (0.0%)	0 (0.0%)
CDRSB	13 (0.8%)	0 (0.0%)
PTEDUCAT	0 (0.0%)	0 (0.0%)
Hippocampus	409 (23.9%)	0 (0.0%)
Ventricles	106 (6.2%)	0 (0.0%)
WholeBrain	82 (4.8%)	0 (0.0%)
ICV	54 (3.2%)	0 (0.0%)
APOE4	0 (0.0%)	0 (0.0%)
Complete Subjects	321/580 (55.3%)	580/580 (100.0%)

Figure 2: Comparison of unimputed and MICE imputed feature completeness.

Feature Engineering

Each model in this study used two distinct input streams, raw longitudinal MRI scans and nine tabular clinical features. This late fusion design reflects the multimodal approach established in the literature, where no single data source captures the full complexity of AD pathology (Venugopalan et al., 2021). No derived features were created.

Raw T1-weighted NIfTI MRI scans were downloaded from the ADNI repository for each subject visit and used as the imaging input stream, with preprocessing and architectural handling described in the Model Architecture section. The MRI stream captures neuroanatomical changes over time without manual feature extraction.

Nine clinical features were selected from the ADNI dataset for use as the tabular input to the Transformer encoder component of each model. Features were chosen based on their established role in the Alzheimer's disease classification literature.

CDRSB and MMSE were selected as the primary cognitive assessment features. CDRSB is a direct measure of dementia severity and has been consistently identified as the most discriminative feature in ADNI-based classification studies, ranking first across multiple independent feature selection methods (Bhagwat et al., 2018). MMSE provides a complementary measure of general cognitive function and has been widely used as a staging indicator for AD (Porsteinsson et al., 2021). Hippocampus volume, Ventricles, WholeBrain, and ICV were included as MRI-derived volumetric measures. Hippocampal atrophy is one of the earliest and most reliable neuroanatomical markers of AD progression, with volumetric decline strongly predictive of MCI-to-AD conversion (Jack et al., 2018). Ventricular enlargement and whole brain volume provide complementary measures of global neurodegeneration (Song et al., 2021). ICV was included as

a normalisation reference, as raw volumetric measures are known to vary with head size. Age and PTEDUCAT were included as demographic variables, reflecting the well-established relationships between age, educational attainment, and cognitive reserve in AD risk stratification. Finally, APOE4 was included as a binary genetic risk indicator, as carrying the APOE ϵ 4 allele is the strongest known genetic risk factor for late-onset AD and has been shown to independently influence MCI conversion rates (Gray et al., 2013).

Train/Validation/Test Split

The 580 subjects were divided into training, validation, and test sets using a 70/15/15 subject-wise split, producing 406 training, 87 validation, and 87 test subjects. This subject-level partitioning is critical in longitudinal studies to ensure that the models are evaluated on entirely unseen individuals rather than recognising familiar subjects from different time points (Aghili, Tabarestani and Adjouadi, 2022). Stratified sampling was applied at both split stages to preserve the class distribution of CN, MCI, and Dementia across all three sets, ensuring that the minority class was adequately represented. To ensure that features with different scales contributed equally to the model, all numerical features were standardised using a z-score normalisation (StandardScaler) to achieve a zero mean and unit standard deviation (El-Sappagh et al., 2021). The scaler was fitted exclusively on training data after splitting and applied independently to validation and test sets to prevent data leakage (Saleh et al., 2023).

Model Architecture

Pseudocode

Simple ANN

```
Algorithm: Simple ANN Training, Evaluation and LIME

INPUT: longitudinal_sequences (ADNI), baseline visit only, clinical_features = 9
OUTPUT: trained model, test metrics, LIME importance scores

INITIALISATION
  Set random seed = 42 for reproducibility
  Partition sequences → train (70%) / val (15%) / test (15%) via stratified split
  Extract baseline visit [0] only per subject (no longitudinal sequence)
  Fit StandardScaler on train baseline clinical features only
  Apply scaler to val and test sets (prevent data leakage)
  Compute class weights from train distribution → alpha for Focal Loss

MODEL ARCHITECTURE
  FOR each subject:
    Load baseline NIfTI MRI scan → normalise [0,1] → resize to 64x64x64
    Extract 9 clinical features from baseline visit only

    MRI stream:
      CNN_encoder(scan) → 128-d MRI embedding

    Fusion:
      concat(128-d MRI, 9-d clinical) → 137-d fused vector
      FC(137+128) → ReLU → Dropout(0.3)
      FC(128+64) → ReLU → Dropout(0.3)
      FC(64+3) → logits [CN, MCI, Dementia]

  NOTE: No temporal processing – each subject represented by single visit only

TRAINING LOOP
  FOR epoch = 1 TO 50:
    FOR each mini-batch (size 4):
      forward_pass(baseline_mri, baseline_clinical) → logits
      loss = FocalLoss(logits, targets, alpha=class_weights, gamma=2.0)
      IF loss is NaN or Inf: skip batch
      loss.backward()
      clip_grad_norm(max_norm = 5.0)
      Adam.step(lr = 1e-4)

    val_loss = evaluate(val_loader)
    IF val_loss < best_val_loss:
      save_checkpoint(model_weights)
      patience_counter = 0
    ELSE:
      patience_counter += 1
    IF patience_counter == 10: BREAK
    IF no val_loss improvement for 5 epochs:
      lr = lr × 0.5

EVALUATION
  Load best checkpoint → evaluate on held-out test set (87 subjects)
  Compute: accuracy, balanced accuracy, macro-F1, weighted-F1,
           per-class precision / recall / F1 / specificity,
           one-vs-rest AUC per class, macro-AUC, test loss

LIME EXPLAINABILITY
  Fit LimeTabularExplainer on train baseline clinical features
  FOR each of 20 test subjects:
    extract baseline clinical features (no visit replication needed)
    perturb clinical values → 40 neighbourhood samples
    fit local linear approximation → extract top-5 feature weights
  Average absolute weights across 20 subjects → mean importance per feature
  Rank features and plot importance chart
```

Figure 3: Pseudocode for the Simple ANN baseline.

TCN-Transformer

```
Algorithm: TCN-Transformer Training, Evaluation and LIME

INPUT: longitudinal_sequences (ADNI), max_visits = 4, clinical_features = 9
OUTPUT: trained model, test metrics, LIME importance scores

INITIALISATION
    Set random seed = 42 for reproducibility
    Partition sequences → train (70%) / val (15%) / test (15%) via stratified split
    Fit StandardScaler on train clinical features only
    Apply scaler to val and test sets (prevent data leakage)
    Compute class weights from train distribution → alpha for Focal Loss

MODEL ARCHITECTURE
    FOR each subject sequence (up to 4 visits):
        Load NIfTI MRI scan → normalise [0,1] → resize to 64×64×64
        Build padding mask: mask[v] = 1 if visit v exists, else 0

        MRI stream:
            CNN_encoder(scan) → 128-d embedding per visit
            transpose embeddings → apply 4 TCN layers in parallel:
                Layer 1: Conv1d(dilation=1) → BatchNorm → ReLU → Dropout(0.2)
                Layer 2: Conv1d(dilation=2) → BatchNorm → ReLU → Dropout(0.2)
                Layer 3: Conv1d(dilation=4) → BatchNorm → ReLU → Dropout(0.2)
                Layer 4: Conv1d(dilation=8) → BatchNorm → ReLU → Dropout(0.2)
            + residual connection at each layer where channel dims match
            Apply padding mask → masked mean pooling → 128-d MRI vector
            NOTE: masking applied at aggregation only; padded timesteps may
                  still influence intermediate convolutional representations

        Clinical stream:
            Linear(9 → 32) + learnable positional encoding
            TransformerEncoder(4 layers, 4 heads, padding_mask) → mean pool → 64-d clinical vector

        Fusion:
            concat(128-d, 64-d) → 192-d fused vector
            FC(192→128) → ReLU → Dropout(0.3)
            FC(128→64) → ReLU → Dropout(0.3)
            FC(64→3) → logits [CN, MCI, Dementia]

TRAINING LOOP
    FOR epoch = 1 TO 50:
        FOR each mini-batch (size 4):
            forward_pass(mri_sequence, clinical_sequence, mask) → logits
            loss = FocalLoss(logits, targets, alpha=class_weights, gamma=2.0)
            IF loss is NaN or Inf: skip batch
            loss.backward()
            clip_grad_norm(max_norm = 5.0)
            Adam.step(lr = 1e-4)

        val_loss = evaluate(val_loader)
        IF val_loss < best_val_loss:
            save_checkpoint(model_weights)
            patience_counter = 0
        ELSE:
            patience_counter += 1
        IF patience_counter == 10: BREAK
        IF no val_loss improvement for 5 epochs:
            lr = lr × 0.5

EVALUATION
    Load best checkpoint → evaluate on held-out test set (87 subjects)
    Compute: accuracy, balanced accuracy, macro-F1, weighted-F1,
             per-class precision / recall / F1 / specificity,
             one-vs-rest AUC per class, macro-AUC, test loss

LIME EXPLAINABILITY
    Fit LimeTabularExplainer on train baseline clinical features
    FOR each of 20 test subjects:
        replicate baseline visit + all 4 visit slots (full attention mask)
        perturb clinical values + 40 neighbourhood samples
        fit local linear approximation → extract top-5 feature weights
    Average absolute weights across 20 subjects → mean importance per feature
    Rank features and plot importance chart
```

Figure 4: Pseudocode for the TCN-Transformer..

BiLSTM-Transformer

```
Algorithm: BiLSTM-Transformer Training and Evaluation

INPUT: longitudinal_sequences (ADNI), max_visits = 4, clinical_features = 9
OUTPUT: trained model, test metrics, LIME importance scores

INITIALISATION
    Set random seed = 42 for reproducibility
    Partition sequences → train (70%) / val (15%) / test (15%) via stratified split
    Fit StandardScaler on train clinical features only
    Apply scaler to val and test sets (prevent data leakage)
    Compute class weights from train distribution → alpha for Focal Loss

MODEL ARCHITECTURE
    FOR each subject sequence (up to 4 visits):
        Load NIFTI MRI scan → normalise [0,1] → resize to 64×64×64
        Build padding mask: mask[v] = 1 if visit v exists, else 0

    MRI stream:
        CNN_encoder(scan) → 128-d embedding per visit
        pack_padded_sequence(embeddings, lengths)
        BiLSTM(packed) → h_forward ⊕ h_backward → 256-d MRI vector

    Clinical stream:
        Linear(9 → 32) + learnable positional encoding
        TransformerEncoder(4 layers, 4 heads, padding_mask) → mean pool → 64-d clinical vector

    Fusion:
        concat(256-d, 64-d) → 320-d fused vector
        FC(320→128) → ReLU → Dropout(0.3)
        FC(128→64) → ReLU → Dropout(0.3)
        FC(64→3) → logits [CN, MCI, Dementia]

TRAINING LOOP
    FOR epoch = 1 TO 50:
        FOR each mini-batch (size 4):
            forward_pass(mri_sequence, clinical_sequence, mask) → logits
            loss = FocalLoss(logits, targets, alpha=class_weights, gamma=2.0)
            loss.backward()
            clip_grad_norm(max_norm = 5.0)
            Adam.step(lr = 1e-4)

        val_loss = evaluate(val_loader)
        IF val_loss < best_val_loss:
            save_checkpoint(model_weights)
            patience_counter = 0
        ELSE:
            patience_counter += 1
        IF patience_counter == 10: BREAK
        IF no val_loss improvement for 5 epochs:
            lr = lr × 0.5

EVALUATION
    Load best checkpoint → evaluate on held-out test set (87 subjects)
    Compute: accuracy, balanced accuracy, macro-F1, weighted-F1,
            per-class precision / recall / F1 / specificity,
            one-vs-rest AUC per class, macro-AUC, test loss

LIME EXPLAINABILITY
    Fit LimeTabularExplainer on train baseline clinical features
    FOR each of 20 test subjects:
        replicate baseline visit → all 4 visit slots (full attention mask)
        perturb clinical values → 40 neighbourhood samples
        fit local linear approximation → extract top-5 feature weights
    Average absolute weights across 20 subjects → mean importance per feature
    Rank features and plot importance chart
```

Figure 5: Pseudocode for the BiLSTM-Transformer.

Shared Themes

All three models shared a common MRI preprocessing pipeline and training configuration to ensure that architectural differences drove any performance differences observed. Raw NIFTI MRI scans were loaded using NiBabel, with NaN values replaced with zeros. Each scan was normalised to the range 0-1 using per-scan min-max scaling and resized to a standardised resolution of 64×64×64 voxels (Suk, Lee and Shen, 2014). All three models processed these standardised volumes through an identical 3D CNN encoder adapted from the framework of Dyrba et al. (2021), shown in Figure 6, comprising three convolutional blocks, each consisting of

a Conv3d layer with kernel size 3 and padding 1, followed by batch normalisation, ReLU activation, and max pooling. Filter sizes progressed from 32 to 64 to 128 across the three blocks, with a global average pooling layer and a fully connected projection producing a 128-dimensional embedding per visit. This shared encoder ensured that differences in downstream performance reflected the temporal modelling capacity of each architecture rather than differences in image feature extraction.

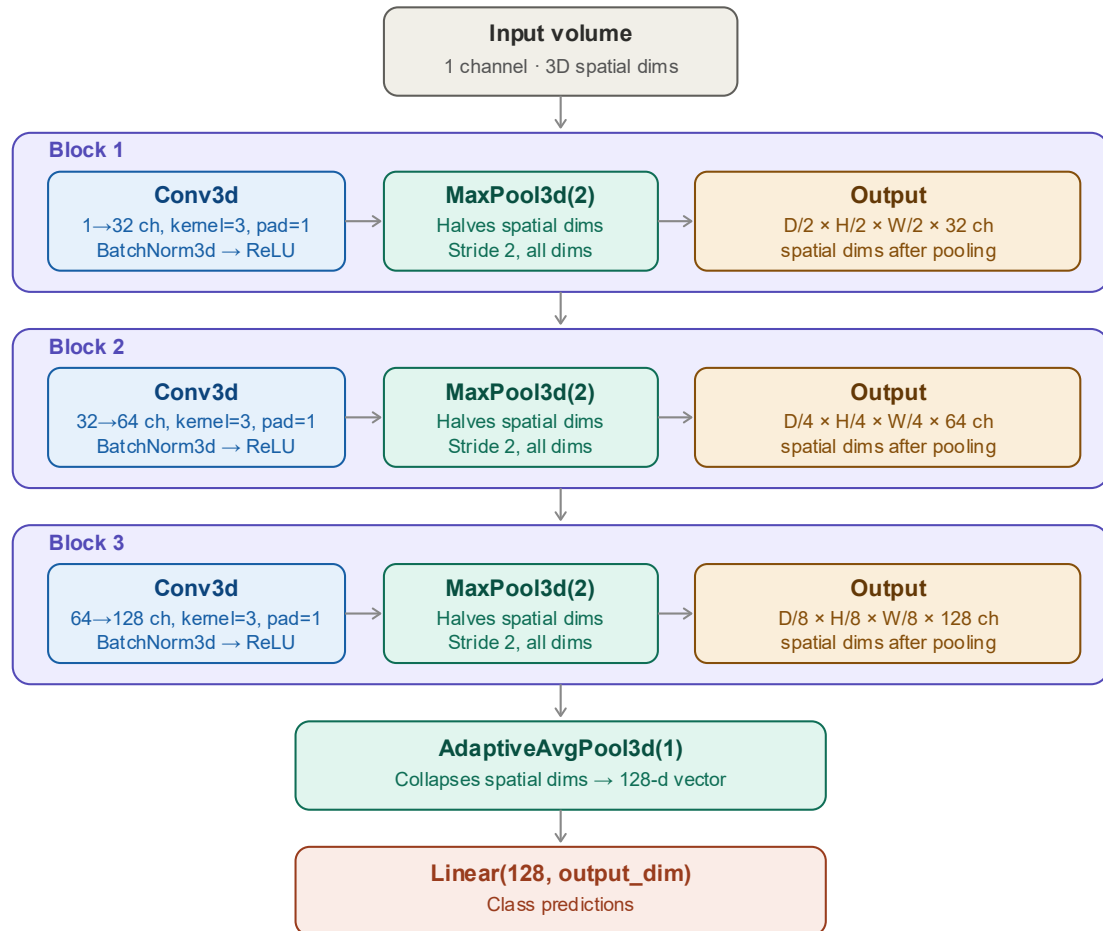


Figure 6: Architecture of the shared 3D CNN encoder, comprising three convolutional blocks with batch normalisation, ReLU activation, and max pooling, producing a 128-dimensional embedding per visit.

All three models were trained under identical conditions. Focal Loss (Lin et al., 2018) was used as the loss function with a gamma value of 2.0, which down-weights easy examples and focuses training on harder misclassified cases. Class weights were calculated from the training set distribution and passed as the alpha parameter to the Focal Loss function, providing an additional mechanism to address the imbalance between CN, MCI, and Dementia classes. The Adam optimiser was used with a learning rate of 1e-4, alongside a ReduceLROnPlateau scheduler that halved the learning rate when validation loss failed to improve for five consecutive epochs. Early stopping with a patience of ten epochs was applied to prevent overfitting, with model weights saved at the checkpoint of best validation loss. Gradient clipping at a maximum norm of 5.0 was applied during backpropagation to prevent exploding gradients (Pascanu, Mikolov and Bengio, 2013). A batch size of four was used throughout, reflecting the memory constraints of loading full 3D MRI volumes per visit. A random seed of 42 was set across all scripts to ensure reproducibility.

Simple ANN

The Simple ANN served as the non-temporal baseline, processing only the baseline visit per subject rather than the full longitudinal sequence. The architecture comprised two input streams fed into a shared fusion classifier, as illustrated in Figure 7. The shared 3D CNN encoder produced a 128-dimensional MRI embedding, which was concatenated with the 9 clinical features to form a single fused representation. This fused vector was passed through a three-layer fully connected classifier with layer sizes 128, 64, and 3, with ReLU activations and dropout regularisation applied after each of the first two layers at rates of 0.3 and 0.3 respectively, following established practices for reducing overfitting in small medical imaging datasets (Srivastava et al., 2014).

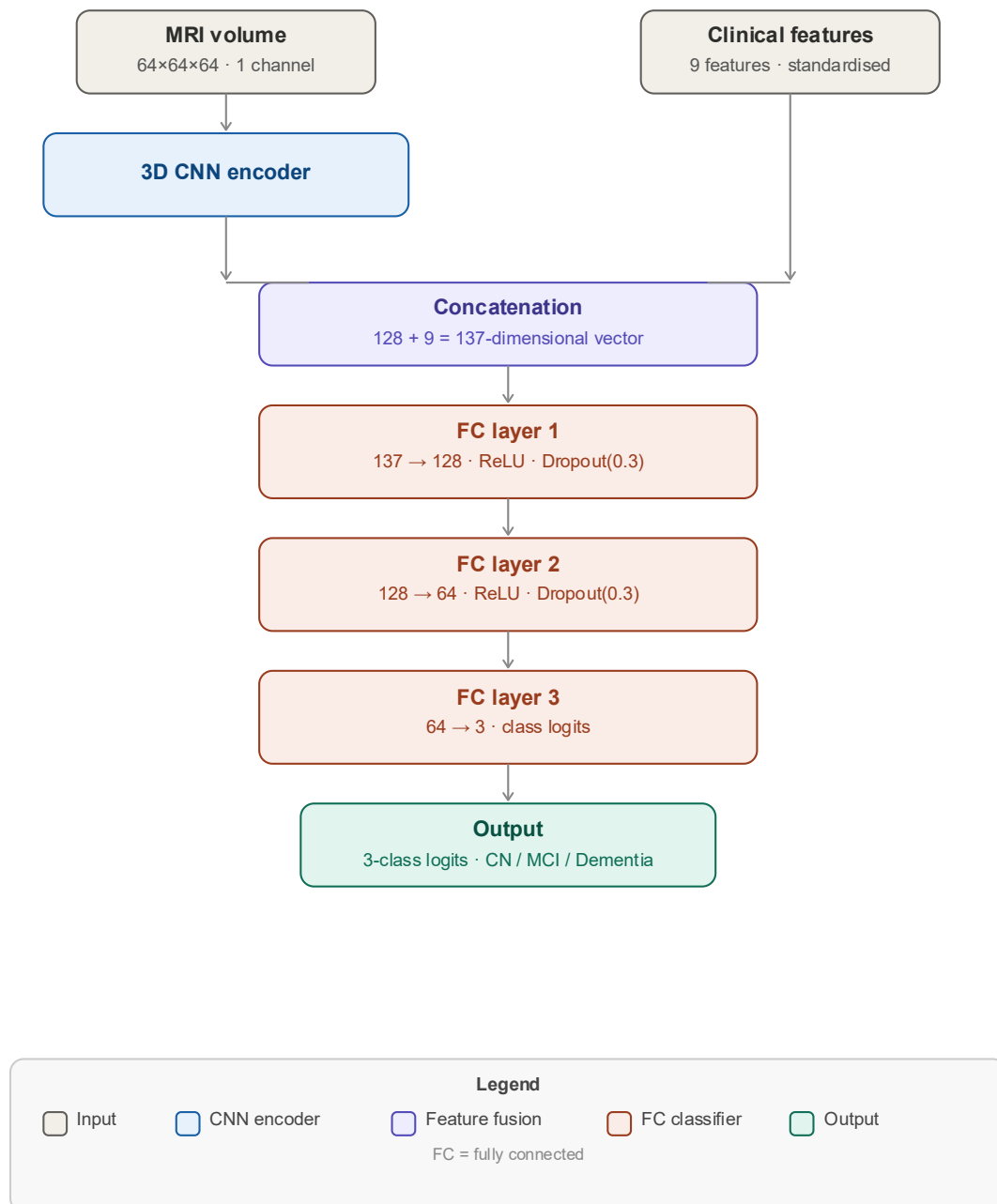


Figure 7: Architecture of the Simple ANN baseline, comprising two input streams, a shared 3D CNN encoder and nine clinical features, concatenated and passed through a three-layer fully connected classifier (128→64→3).

The Simple ANN was selected as the baseline because it processes each visit in isolation with no temporal awareness, making it directly comparable to conventional cross-sectional approaches in the AD classification literature (Bhagwat et al., 2018). Any performance improvement observed in the BiLSTM-Transformer and TCN-Transformer architectures over this baseline can subsequently be attributed to their capacity to model longitudinal disease progression rather than differences in feature extraction or fusion strategy.

TCN-Transformer

The TCN-Transformer processed the full longitudinal sequence of up to four visits per subject using two parallel temporal streams, as illustrated in Figure 8. This dual-stream design reflects the complementary nature of the MRI and clinical modalities, with each stream using a temporal architecture suited to its input structure before fusion for final classification.

In the MRI stream, the shared 3D CNN encoder extracted a 128-dimensional embedding per visit, producing a sequence of visit-level embeddings that was passed into a four-layer Temporal Convolutional Network. Each TCN layer applied dilated convolutions with exponentially increasing dilation rates of 1, 2, 4, and 8, a kernel size of 3, and a hidden size of 128, followed by batch normalisation, ReLU activation, and dropout at a rate of 0.2. The exponentially increasing dilation rates allow the receptive field to grow across the visit sequence without increasing the number of parameters, enabling the model to capture both short-term visit-to-visit changes and longer-range longitudinal trends simultaneously (Bai, Kolter and Koltun, 2018). Residual connections were applied between layers where channel dimensions matched, stabilising the gradient flow during training. Masked mean pooling was applied to the TCN output to account for varying sequence lengths across subjects, producing a single 128-dimensional temporal MRI representation. However, masking is only applied during aggregation, meaning padded timesteps may still influence intermediate temporal convolutions.

In the clinical stream, the nine standardised clinical features per visit were linearly projected to a 32-dimensional embedding and combined with a learnable positional encoding to preserve visit ordering. These embeddings were then processed through a four-layer Transformer encoder with four attention heads, allowing the model to weight the relative importance of different visits dynamically rather than treating all timepoints equally (Vaswani et al., 2017). The output sequence was mean-pooled and projected to a 64-dimensional clinical representation.

The two stream outputs were concatenated to form a 192-dimensional fused vector, which was passed through a three-layer fully connected classifier with layer sizes 128, 64, and 3, with ReLU activations and dropout of 0.3 after each of the first two layers, producing the final three-class prediction over CN, MCI, and Dementia.

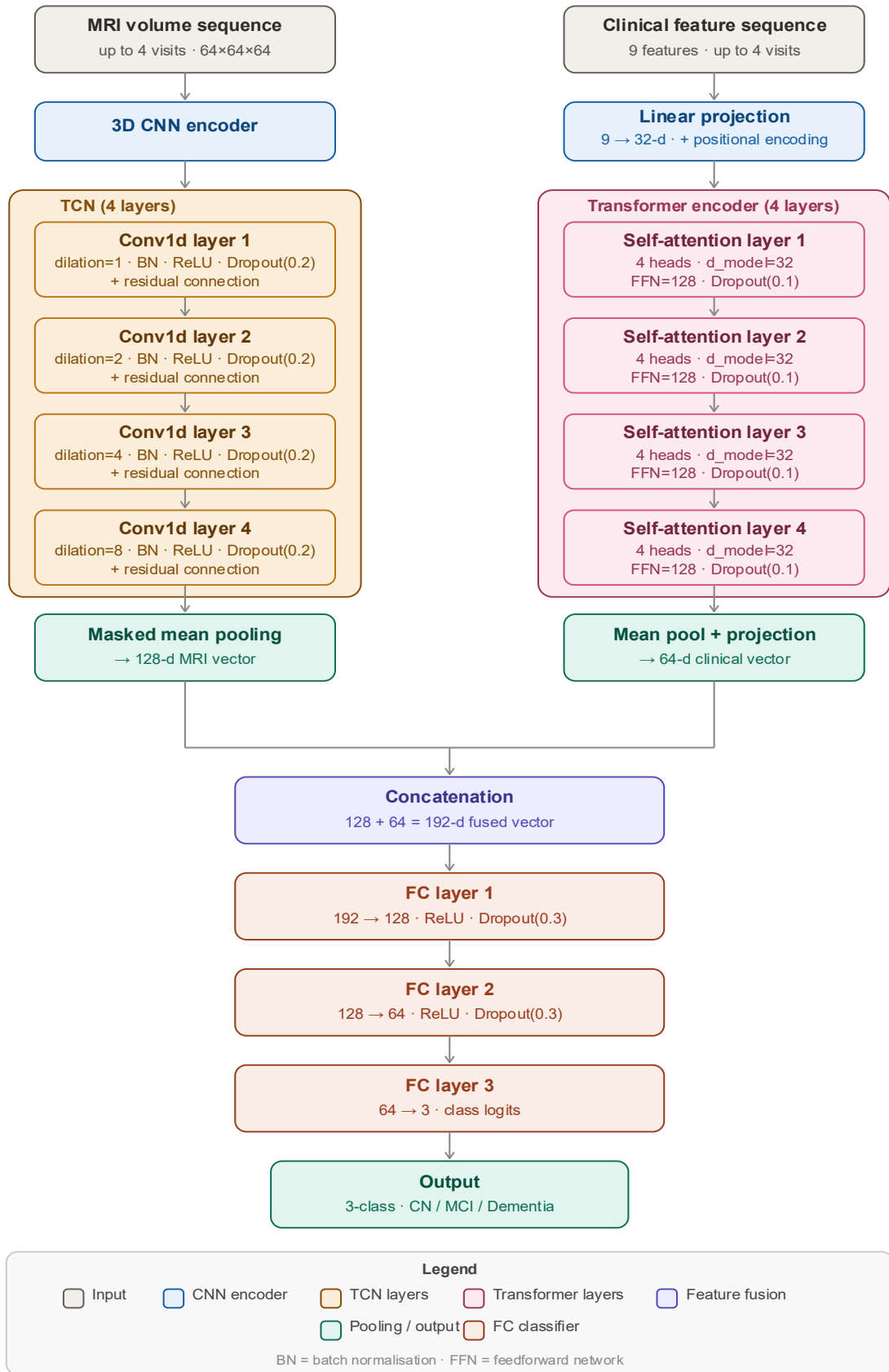


Figure 8: Architecture of the TCN-Transformer, comprising parallel MRI and clinical temporal streams. The MRI stream processes visit embeddings through a four-layer TCN with dilated convolutions; the clinical stream uses a four-layer Transformer encoder. Outputs are concatenated and passed through a three-layer classifier (128→64→3).

BiLSTM-Transformer

The BiLSTM-Transformer processed the full longitudinal sequence of up to four visits per subject using two parallel temporal streams, as illustrated in Figure 9. The architecture mirrors the dual-stream design of the TCN-Transformer, with the key distinction that the MRI stream employs a BiLSTM network rather than dilated convolutions to model temporal dependencies across visits.

In the MRI stream, the shared 3D CNN encoder extracted a 128-dimensional embedding per visit, producing a sequence of visit-level embeddings that was passed into a two-layer BiLSTM with a hidden size of 128 and dropout of 0.2 between layers. Variable-length sequences were handled using packed sequence processing, which ensures padded visits are excluded from the LSTM computation during training, providing more robust handling of missing visits than the aggregation-stage masking used in the TCN stream. The BiLSTM processed the sequence in both forward and backward directions simultaneously, allowing later visits to contextualise earlier measurements and vice versa. The final hidden states from the last layer of each direction were concatenated, producing a 256-dimensional temporal MRI representation that encodes the full longitudinal trajectory of neuroanatomical change across all visits.

The clinical stream was identical to the TCN-Transformer, with the nine standardised clinical features per visit linearly projected to a 32-dimensional embedding, combined with a learnable positional encoding, and processed through a four-layer Transformer encoder with four attention heads and explicit padding masks applied during training. The resulting sequence was mean-pooled and projected to a 64-dimensional clinical representation.

The two stream outputs were concatenated to form a 320-dimensional fused vector, larger than the TCN-Transformer's 192-dimensional fusion due to the BiLSTM's bidirectional output, which was passed through a three-layer fully connected classifier with layer sizes 128, 64, and 3, with ReLU activations and dropout of 0.3 after each of the first two layers, producing the final three-class prediction for CN, MCI, and Dementia.

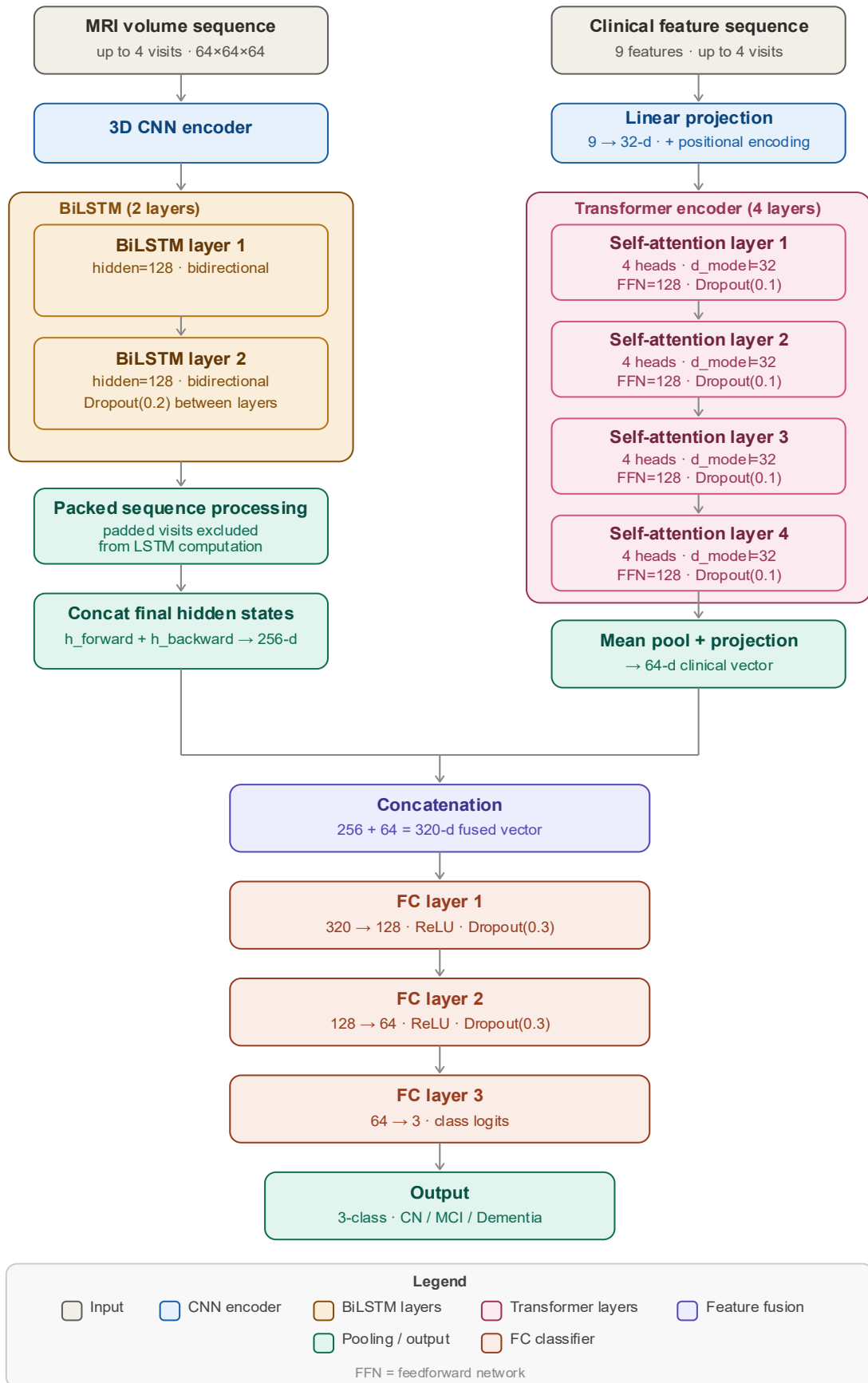


Figure 9: Architecture of the BiLSTM-Transformer, comprising parallel MRI and clinical temporal streams. The MRI stream processes visit embeddings through a two-layer BiLSTM; the clinical stream uses a four-layer Transformer encoder. Outputs are concatenated and passed through a three-layer classifier ($128 \rightarrow 64 \rightarrow 3$).

LIME Explainability Analysis

Following training, LIME was applied to each of the three trained models to identify the clinical features most influential in driving individual predictions (Ribeiro et al., 2016). A LimeTabularExplainer was fitted using the standardised baseline visit clinical features from the training set, providing LIME with a reference over which to generate perturbations. For each explanation, LIME generated 40 perturbed samples in the neighbourhood of the input instance and fitted a locally linear approximation to identify the five most influential features for the predicted class.

Explanations were generated for 20 test set subjects per model, with feature importance scores extracted as signed weights from the local approximation. Absolute values were taken to reflect the magnitude of each feature's influence regardless of direction, and scores were averaged across all 20 subjects to produce a single mean importance score per feature per model. This aggregation strategy allows cross-model comparison of feature importance hierarchies rather than relying on any individual prediction.

Across all three models, LIME operates on the clinical stream only, as LimeTabularExplainer requires tabular input, so the MRI scans were not used for LIME analysis. The baseline visit features were replicated across all four visit slots with a full attention mask during inference, meaning LIME probes the model's sensitivity to variation in baseline clinical values rather than longitudinal change.

Ethical Considerations

This study was conducted in accordance with the ethical approval granted by Northumbria University's ethics review process prior to data collection. The neuroimaging and clinical data used in this study were obtained from the ADNI database (adni.loni.usc.edu). Access required formal application describing the intended research purpose, followed by agreement to the ADNI Data Use Agreement, which says that data may only be used for non-commercial research purposes and must not be redistributed. All conditions of the agreement were met throughout this study.

ADNI data is anonymised prior to release, and no personally identifiable information was accessible at any point during this research. No individual patient cases were identified, evaluated, or reported in this study, and all analysis was conducted at the aggregate level. The models developed represent research tools only, no clinical decisions were made or recommended on the basis of model outputs, and the work carries no intent of clinical deployment without further validation.

All data was stored exclusively on a secured personal device and was not uploaded to any external platform, version control system, or cloud storage service at any point during the project. No data is included in the submitted dissertation beyond summary statistics and visualisations that do not permit identification of individual subjects.

Results

Simple ANN Metrics

Imputed Model

Simple ANN - Training Summary

Metric	Value
Total Epochs Trained	43
Best Validation Loss	0.1911
Final Training Accuracy	74.88%
Final Validation Accuracy	74.71%
Test Accuracy	79.31%
Best Training Loss	0.1680
Best Validation Accuracy	79.31%

Figure 10: Training summary for the Simple ANN including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy.

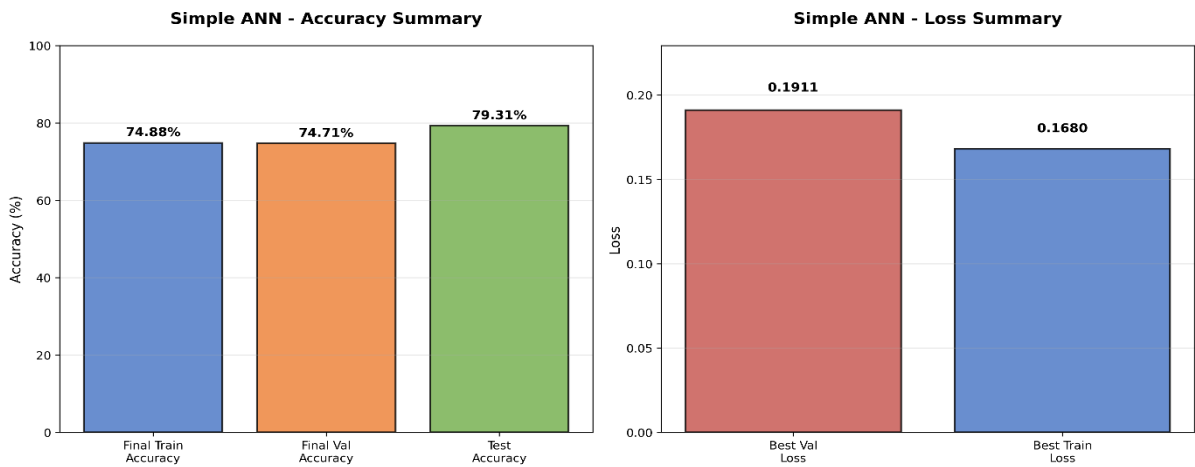


Figure 11: Accuracy and loss summary bar charts for the Simple ANN, showing final training accuracy, final validation accuracy, test accuracy, best validation loss, and best training loss

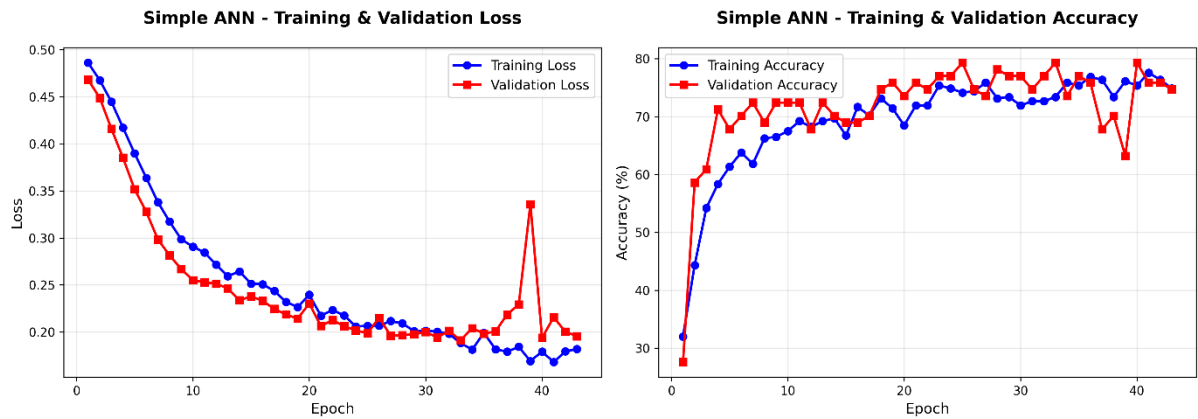


Figure 12: Training and validation loss and accuracy curves for the Simple ANN imputed model across 43 epochs.

The Simple ANN trained on MICE-imputed data converged over 43 epochs before early stopping was triggered. Training and validation loss declined consistently across 43 epochs, with a brief spike in validation loss around epoch 39 that recovered without sustained divergence. Accuracy rose rapidly in the first ten epochs and converged in the upper 70s by the end of training. The best validation loss achieved was 0.1911, with a corresponding best training loss of 0.1680. The final training accuracy was 74.88% and the final validation accuracy was 74.71%, with a best validation accuracy of 79.31%. The model achieved a test accuracy of 79.31%, as shown in Figure 10.

Simple ANN - Training Summary

Metric	Value
Total Epochs Trained	50
Best Validation Loss	0.1883
Final Training Accuracy	79.56%
Final Validation Accuracy	66.67%
Test Accuracy	79.31%
Best Training Loss	0.1625
Best Validation Accuracy	78.16%

Figure 13: Training summary for the unimputed Simple ANN including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy.

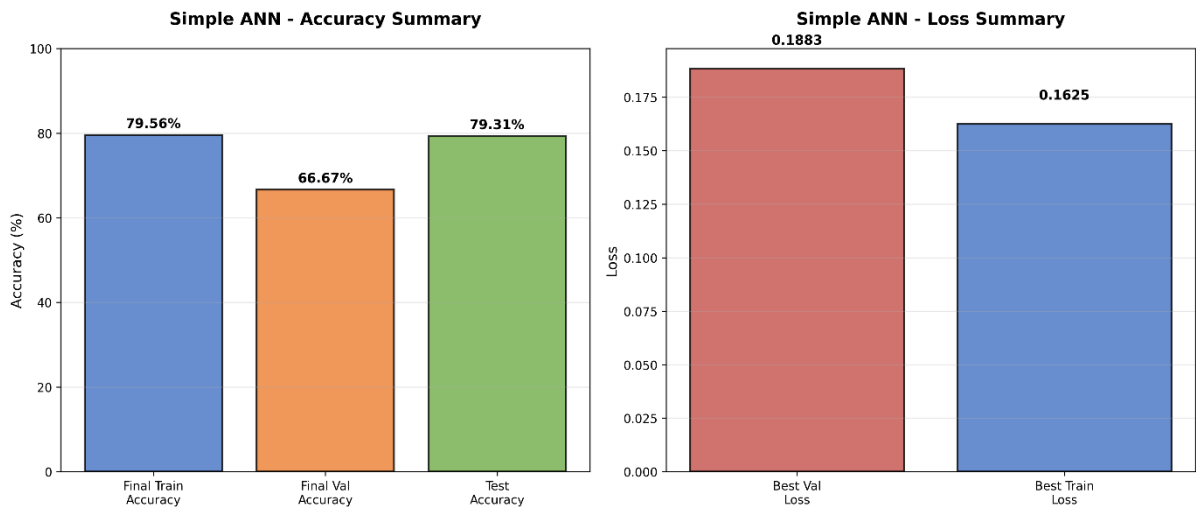


Figure 14: Accuracy and loss summary bar charts for the unimputed Simple ANN, showing final training accuracy, final validation accuracy, test accuracy, best validation loss, and best training loss

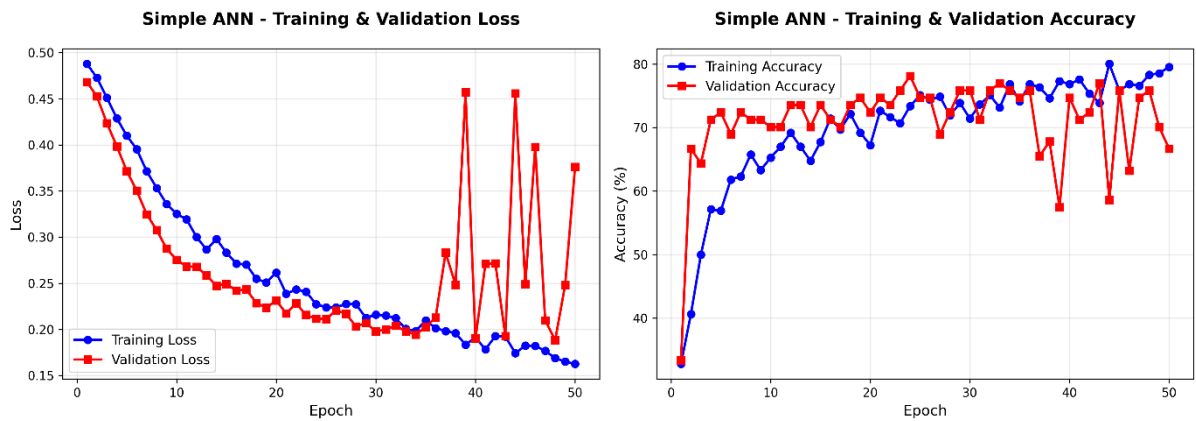


Figure 15: Training and validation loss and accuracy curves for the unimputed Simple ANN imputed model across 50 epochs.

The unimputed Simple ANN trained for the full 50 epochs. The training loss followed a similar trajectory to the imputed version, however validation loss showed considerably more instability in later epochs, with multiple large spikes between epochs 36 and 50. The final validation accuracy was notably lower at 66.67%, reflecting this instability, with a best validation accuracy of 78.16%. The model nonetheless achieved a test accuracy of 79.31%, identical to the imputed version, indicating that MICE imputation provided no measurable accuracy gain for the Simple ANN on this dataset.

BiLSTM-Transformer Metrics

Imputed Model

BiLSTM-Transformer - Training Summary

Metric	Value
Total Epochs Trained	30
Best Validation Loss	0.1385
Final Training Accuracy	85.96%
Final Validation Accuracy	83.91%
Test Accuracy	88.51%
Best Training Loss	0.1110
Best Validation Accuracy	87.36%

Figure 16: Training summary for the BiLSTM-Transformer including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy.

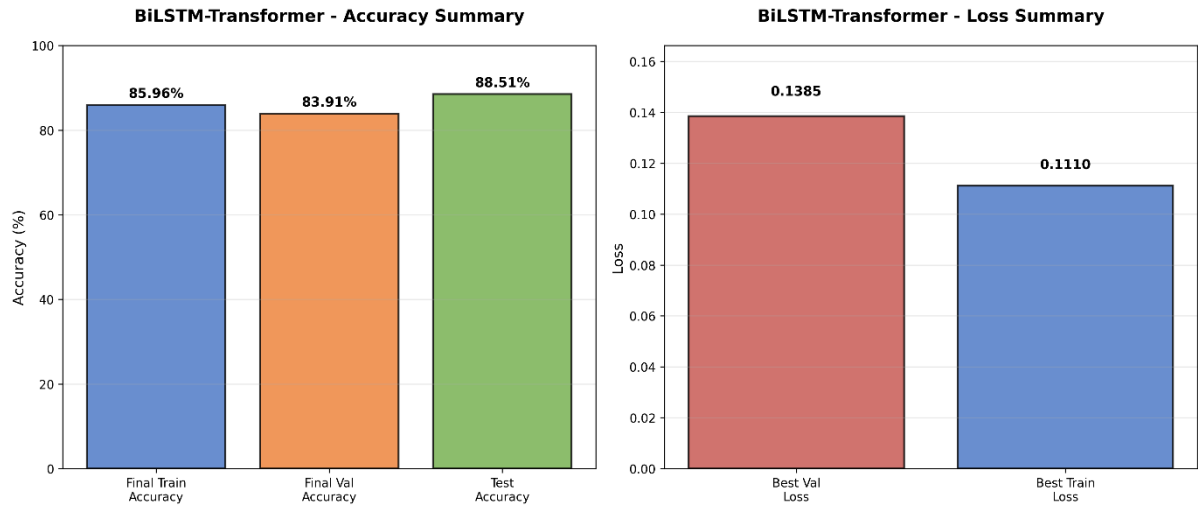


Figure 17: Accuracy and loss summary bar charts for the BiLSTM-Transformer, showing final training accuracy, final validation accuracy, test accuracy, best validation loss, and best training loss

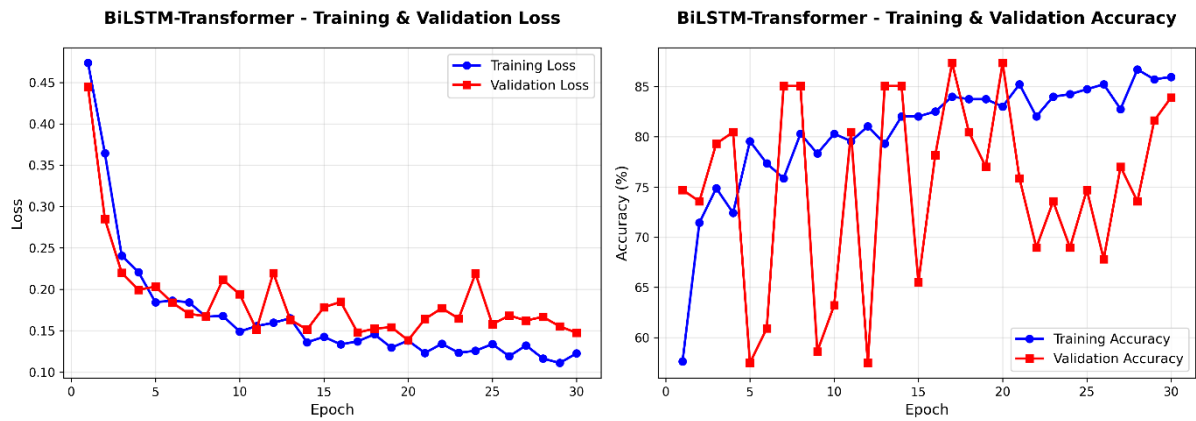


Figure 18: Training and validation loss and accuracy curves for the BiLSTM-Transformer model across 30 epochs.

The BiLSTM-Transformer trained on MICE-imputed data converged over 30 epochs. Training and validation loss declined steadily across 30 epochs. The validation accuracy curve showed large oscillation throughout, fluctuating between approximately 58% and 87%, reflecting the small validation set size where individual misclassifications produce large proportional swings. The best validation loss was 0.1385, with a best training loss of 0.1110. The final training accuracy was 85.96% and the final validation accuracy was 83.91%, with a best validation accuracy of 87.36%. The model achieved a test accuracy of 88.51%, as shown in Figure 16.

BILSTM-Transformer - Training Summary

Metric	Value
Total Epochs Trained	30
Best Validation Loss	0.1365
Final Training Accuracy	84.98%
Final Validation Accuracy	78.16%
Test Accuracy	82.76%
Best Training Loss	0.1162
Best Validation Accuracy	86.21%

Figure 19: Training summary for the unimputed BiLSTM-Transformer including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy

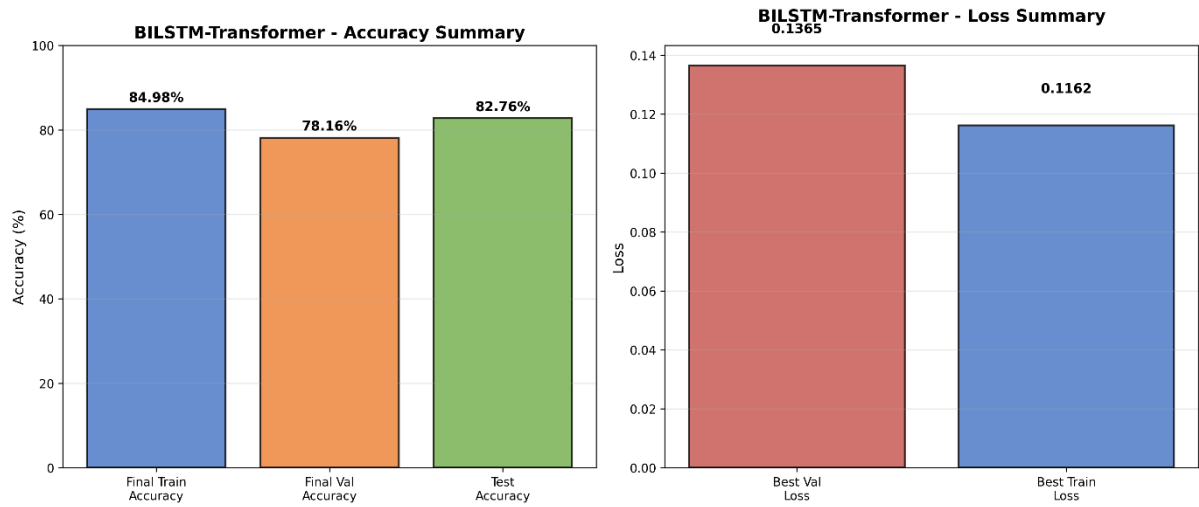


Figure 20: Accuracy and loss summary bar charts for the unimputed BiLSTM-Transformer, showing final training accuracy, final validation accuracy, test accuracy, best validation loss, and best training loss

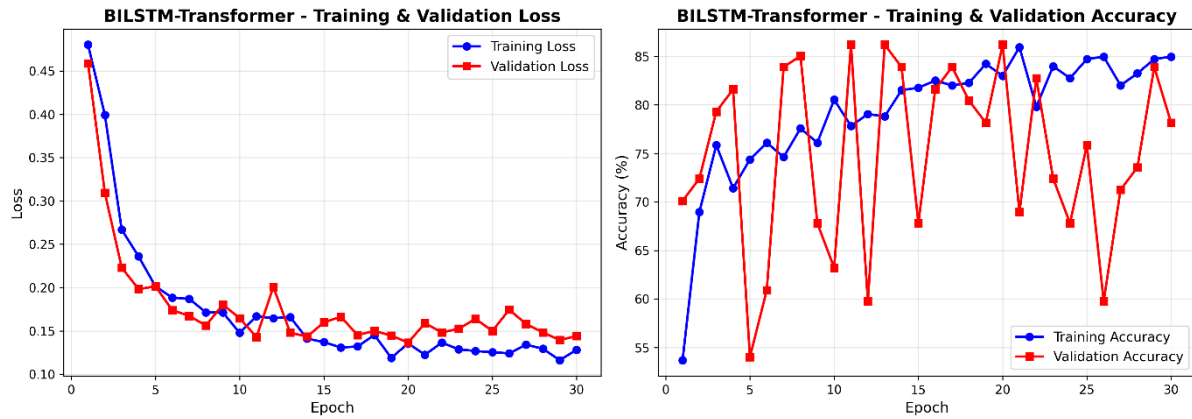


Figure 21: Training and validation loss and accuracy curves for the unimputed BiLSTM-Transformer model across 30 epochs.

The unimputed BiLSTM-Transformer also trained for 30 epochs. The unimputed BiLSTM-Transformer followed a closely comparable training profile to the imputed version, with validation accuracy again showing high variability across training. The best validation loss was 0.1365, marginally lower than the imputed model, with a best training loss of 0.1162. The final training accuracy was 84.98% and the final validation accuracy was 78.16%, with a best validation accuracy of 86.21%. The model achieved a test accuracy of 82.76%, representing a reduction of 5.75 percentage points compared to the imputed version, indicating a meaningful benefit from MICE imputation for this architecture.

TCN-Transformer Metrics

Imputed Model

TCN-Transformer - Training Summary

Metric	Value
Total Epochs Trained	32
Best Validation Loss	0.1372
Final Training Accuracy	83.99%
Final Validation Accuracy	79.31%
Test Accuracy	87.36%
Best Training Loss	0.1101
Best Validation Accuracy	83.91%

Figure 22: Training summary for the TCN-Transformer including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy.

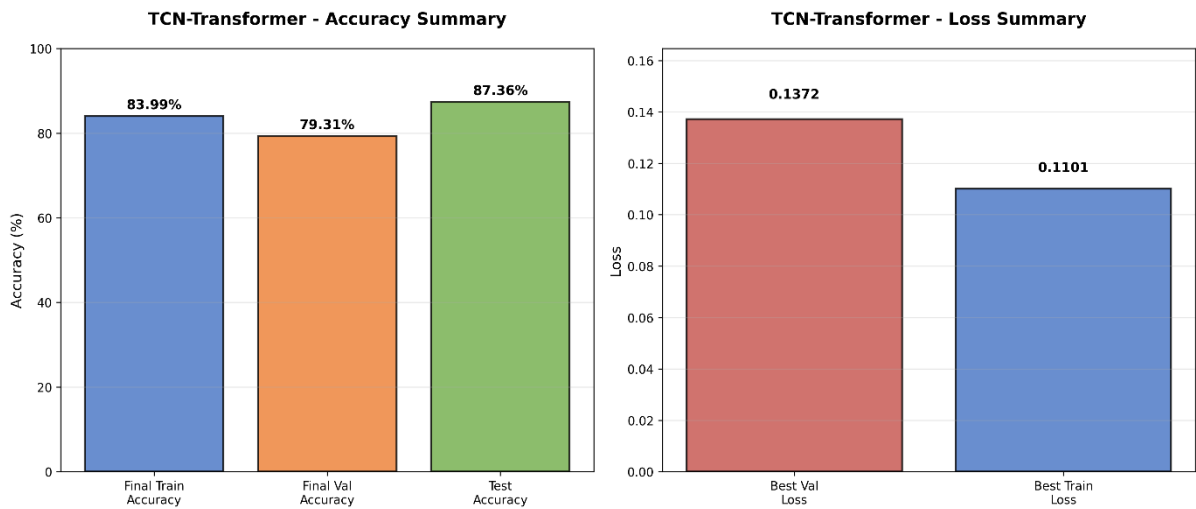


Figure 23: Accuracy and loss summary bar charts for the TCN-Transformer, showing final training accuracy, final validation accuracy, test accuracy, best validation loss, and best training loss.

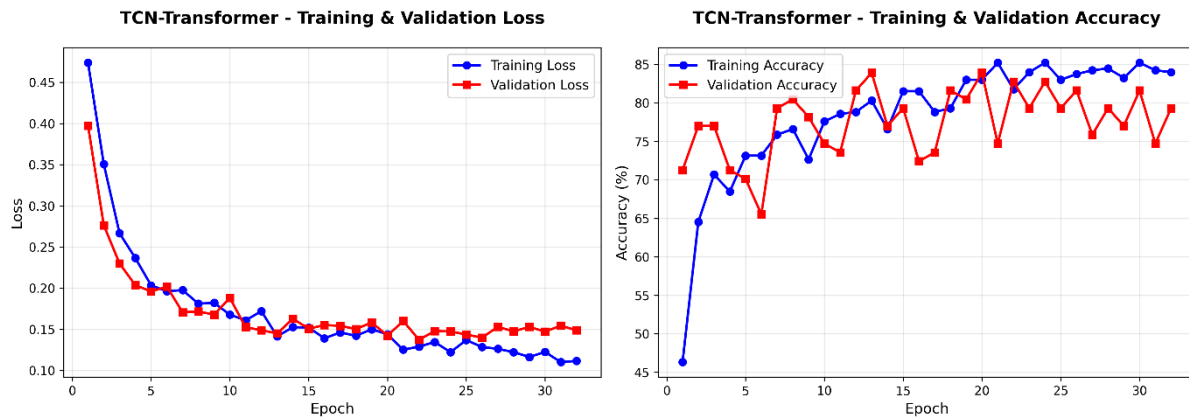


Figure 24: Training and validation loss and accuracy curves for the TCN-Transformer model across 32 epochs.

The TCN-Transformer trained on MICE-imputed data converged over 32 epochs. Training loss declined to around 0.11 by epoch 32, the lowest final training loss of the three models. Validation accuracy showed oscillation similar to the BiLSTM-Transformer but with less extreme variation, ranging between approximately 65% and 84%. The best validation loss was 0.1372, with a best training loss of 0.1101. The final training accuracy was 83.99% and the final validation accuracy was 79.31%, with a best validation accuracy of 83.91%. The model achieved a test accuracy of 87.36%, as shown in Figure 22.

Unimputed Model

TCN-Transformer - Training Summary

Metric	Value
Total Epochs Trained	32
Best Validation Loss	0.1531
Final Training Accuracy	85.96%
Final Validation Accuracy	74.71%
Test Accuracy	85.06%
Best Training Loss	0.1099
Best Validation Accuracy	81.61%

Figure 25: Training and validation loss and accuracy curves for the unimputed TCN-Transformer model across 32 epochs

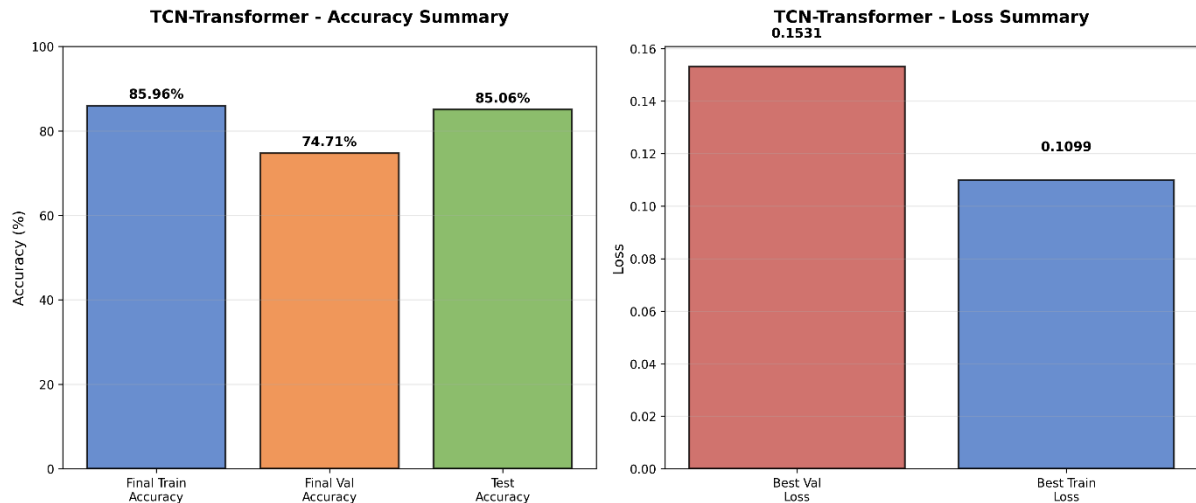


Figure 26: Training summary for the unimputed TCN-Transformer including total epochs trained, best validation loss, final training and validation accuracy, best validation accuracy, and test accuracy

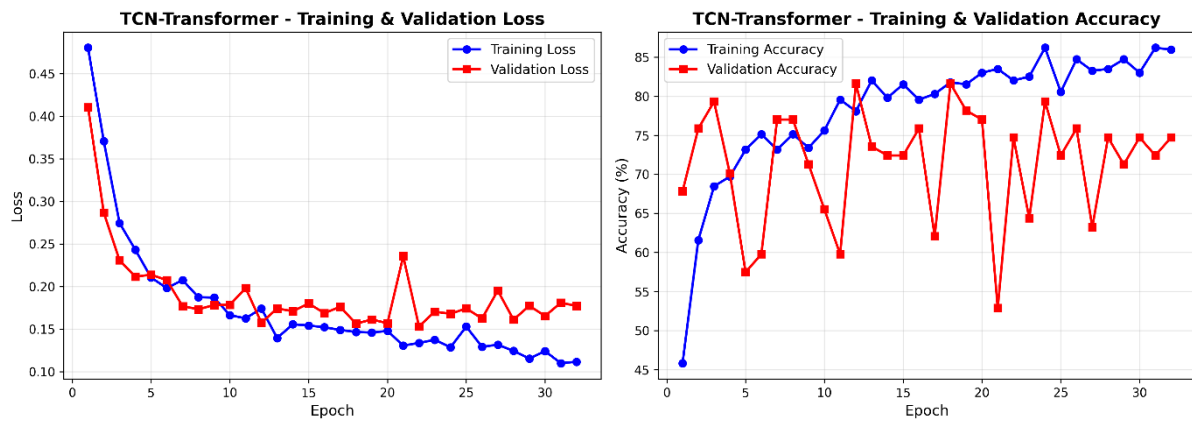


Figure 27: Training and validation loss and accuracy curves for the unimputed TCN-Transformer model across 32 epochs.

The unimputed TCN-Transformer also trained for 32 epochs. Training loss closely mirrored the imputed version. Validation loss stabilised around 0.18 rather than 0.15, and validation accuracy showed the highest oscillation of any model in this study, ranging between approximately 53% and 82% with no visible upward trend in later epochs. The best validation loss was 0.1531, higher than the imputed model's 0.1372, with a best training loss of 0.1099. The final training accuracy was 85.96% and the final validation accuracy was 74.71%, with a best validation accuracy of 81.61%. The model achieved a test accuracy of 85.06%, a reduction of 2.30 percentage points compared to the imputed version, confirming that MICE imputation provided a moderate but consistent improvement for the TCN-Transformer architecture.

Comparison Metrics

Model Performance Comparison			
Metric	Simple ANN	BiLSTM-Transformer	TCN-Transformer
Test Accuracy (%)	79.31	88.51	87.36
Balanced Accuracy (%)	77.59	88.14	85.8
Sensitivity (%)	77.59	88.14	85.8
Specificity (%)	90.14	94.61	94.05
Macro F1	0.773	0.8734	0.8548
Weighted F1	0.7991	0.8866	0.8752
Macro AUC	0.9187	0.9637	0.9676
Test Loss	0.1911	N/A	0.1372

Simple ANN

BiLSTM-Transformer

TCN-Transformer

Figure 28: Model performance comparison across key evaluation metrics for all three imputed models, with the best-performing value per metric highlighted in the corresponding model colour.

Overall Classification Performance

The model performance comparison table summarises the key evaluation metrics for all three imputed models. The BiLSTM-Transformer achieved the highest test accuracy at 88.51%, followed by the TCN-Transformer at 87.36%, with the Simple ANN performing notably lower at 79.31%. The same ordering was observed for balanced accuracy, where the BiLSTM-Transformer scored 88.14%, the TCN-Transformer 85.80%, and the Simple ANN 77.59%. The alignment between accuracy and balanced accuracy across all three models indicated that class imbalance did not substantially distort the overall accuracy figures.

The BiLSTM-Transformer led on macro F1 (0.8734 vs. 0.8548), weighted F1 (0.8866 vs. 0.8752), overall specificity (94.61% vs. 94.05%), and sensitivity, while the TCN-Transformer led marginally on macro-AUC (0.9676 vs. 0.9637) and recorded the lowest test loss at 0.1372 compared to 0.1385 for the BiLSTM-Transformer and 0.1911 for the Simple ANN. These different strengths reflect complementary rather than hierarchical performance between the two temporal architectures. Both temporal models outperformed the Simple ANN across all reported metrics.

Confusion Matrices

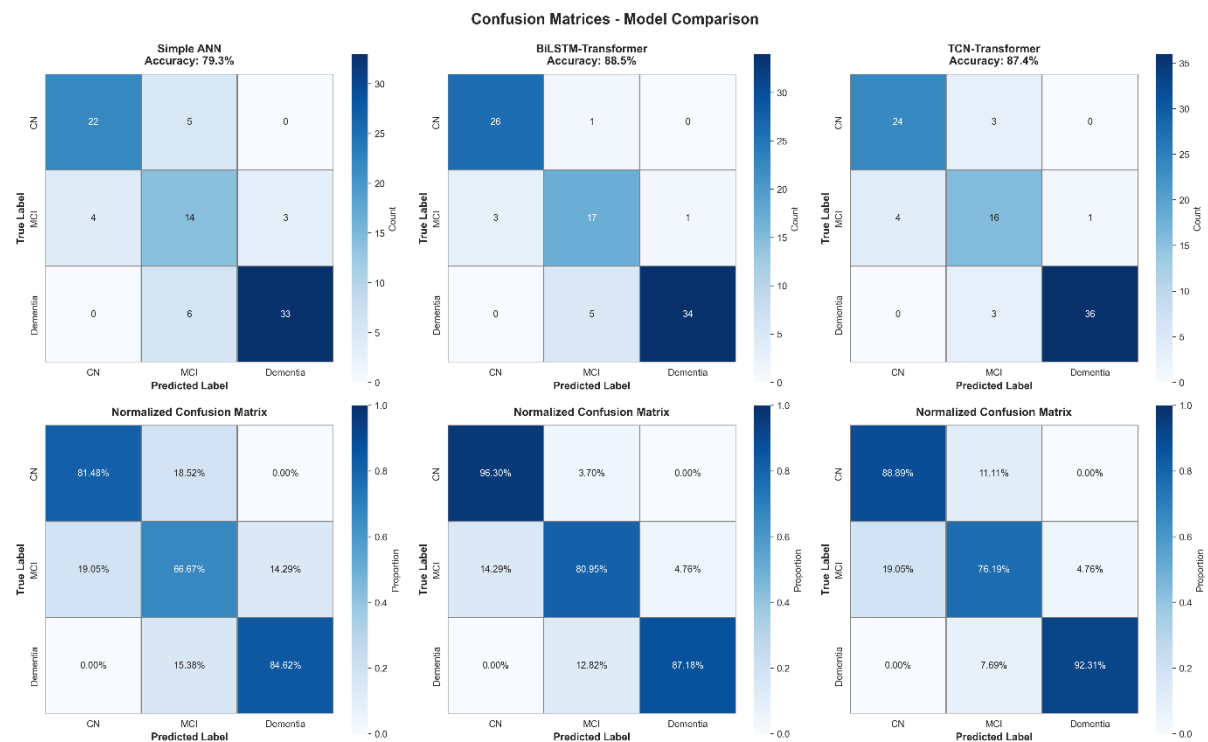


Figure 29: Raw count and normalised confusion matrices for the Simple ANN, BiLSTM-Transformer, and TCN-Transformer on the imputed test set, showing per-class classification performance across CN, MCI, and Dementia.

The confusion matrix figure presents both raw count and normalised confusion matrices for each model. Across all three models, classification of the Dementia class was strongest. The TCN-Transformer correctly classified 36 of 39 Dementia samples (92.31%), the BiLSTM-Transformer correctly classified 34 (87.18%), and the Simple ANN correctly classified 33 (84.62%). In all cases, the only source of Dementia misclassification was with MCI, with no Dementia samples misclassified as CN by any model.

CN classification was also strong across all models. The BiLSTM-Transformer achieved the highest CN recall at 96.30%, correctly classifying 26 of 27 samples. The TCN-Transformer correctly classified 24 (88.89%) and the Simple ANN 22 (81.48%), with CN errors directed exclusively toward MCI in all three cases.

MCI was the most difficult class to classify across all models. The Simple ANN correctly classified 14 of 21 MCI samples (66.67%), with 4 misclassified as CN and 3 as Dementia. The BiLSTM-Transformer improved this to 17 correct (80.95%), and the TCN-Transformer classified 16 correctly (76.19%). The difficulty in classifying MCI samples was consistent across all architectures, which was unsurprising given that MCI represents a transitional clinical position between normal cognition and dementia, making it naturally harder to distinguish from either adjacent class.

ROC Curves

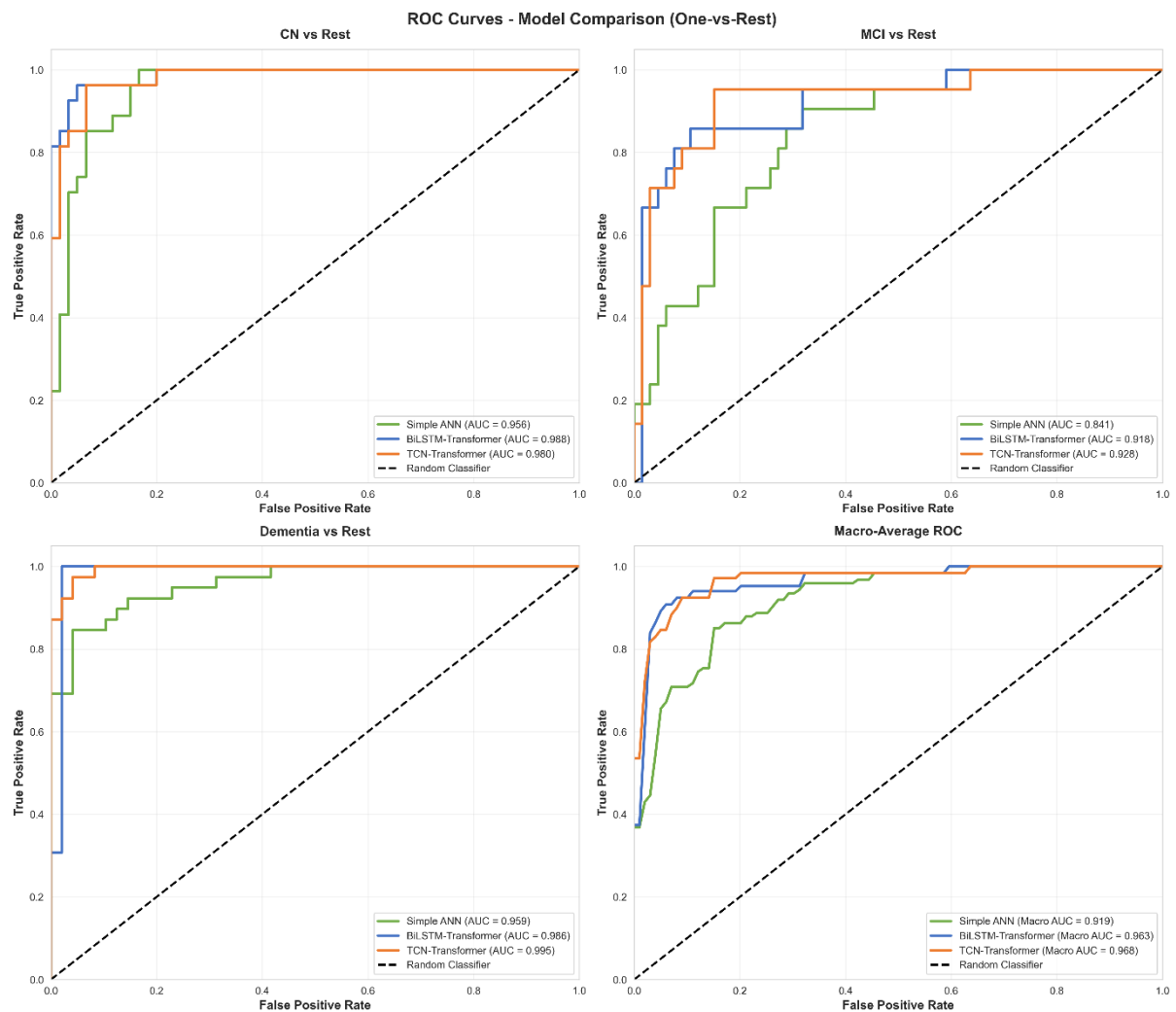


Figure 30: One-vs-rest ROC curves per class and macro-average for all three imputed models.

The ROC curve figure presents one-vs-rest curves for each class alongside the macro-average ROC for all three models. All models substantially exceeded the random classifier baseline across every class. For CN classification, the BiLSTM-Transformer achieved an AUC of 0.988, the TCN-Transformer 0.980, and the Simple ANN 0.956. For MCI, which produced the lowest AUC scores across all models, the TCN-Transformer led with 0.928, followed by the BiLSTM-Transformer at 0.918 and the Simple ANN at 0.841. For Dementia, the TCN-Transformer again achieved the highest AUC of 0.995, with the BiLSTM-Transformer at 0.986 and the Simple ANN at 0.959. The macro-average AUC was 0.968 for the TCN-Transformer, 0.963 for the BiLSTM-Transformer, and 0.919 for the Simple ANN.

Precision-Recall Curves

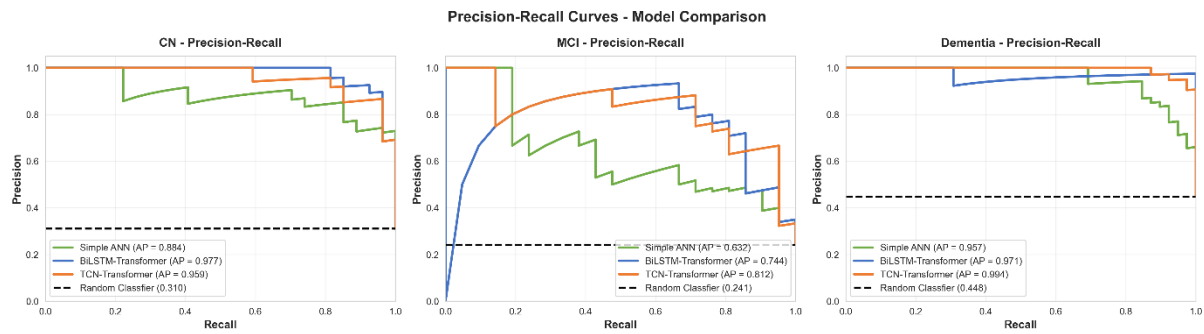


Figure 31: Precision-recall curves per class for all three imputed models, with random classifier baseline shown as a dashed line.

The precision-recall curve figure presents per-class curves for each model, with a random classifier shown as a dashed line at the class prevalence for each class. All three models substantially exceeded the random classifier across every class.

For CN, the BiLSTM-Transformer achieved the highest average precision of 0.977, followed by the TCN-Transformer at 0.959 and the Simple ANN at 0.884. For Dementia, the TCN-Transformer led with an average precision of 0.994, compared to 0.971 for the BiLSTM-Transformer and 0.957 for the Simple ANN. MCI produced the lowest average precision scores across all models, with the TCN-Transformer achieving 0.812, the BiLSTM-Transformer 0.744, and the Simple ANN 0.632. Despite the Simple ANN's MCI average precision of 0.632 appearing low in absolute terms, it remained substantially above the random classifier level of 0.241, indicating that the model retained meaningful discriminative ability for that class. The gap between the temporal models and the Simple ANN was most pronounced for MCI, where the Simple ANN curve showed considerably earlier precision degradation as recall increased.

Per-Class Metrics

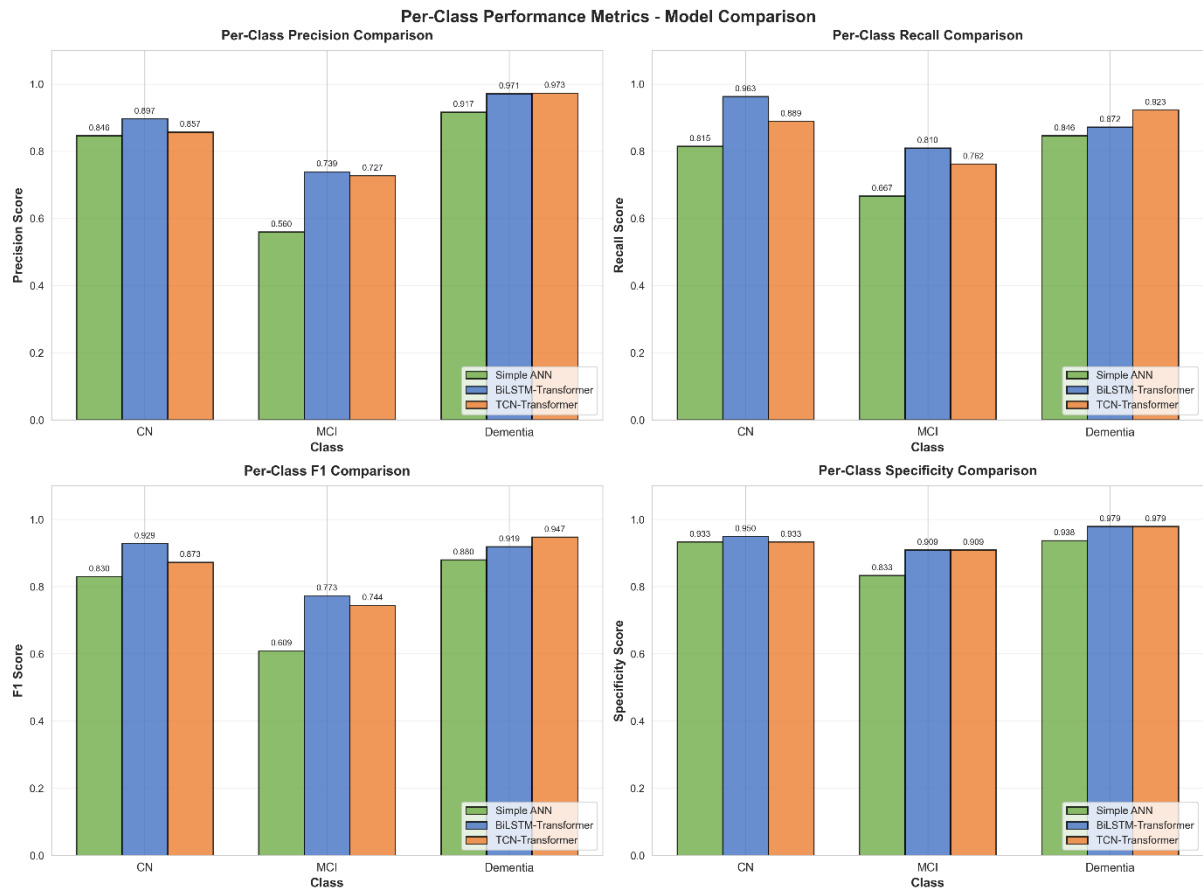


Figure 32: Per-Class precision, recall, F1-score, and specificity for all three imputed models across CN, MCI, and Dementia classes.

The per-class performance metrics figure presents precision, recall, F1-score, and specificity broken down by class for all three models. Dementia precision was highest overall across all architectures, while MCI precision was the lowest, with the Simple ANN's 0.560 notably weaker than the temporal models at 0.739 and 0.727. The same pattern held for recall, with MCI the weakest class for all models and the BiLSTM-Transformer achieving the strongest CN recall at 0.963. MCI F1 was the lowest class score across all models, while Dementia F1 was the highest, reaching 0.947 for the TCN-Transformer. MCI specificity was the lowest in each case, at 0.909 for both temporal models and 0.833 for the Simple ANN.

Prediction Confidence

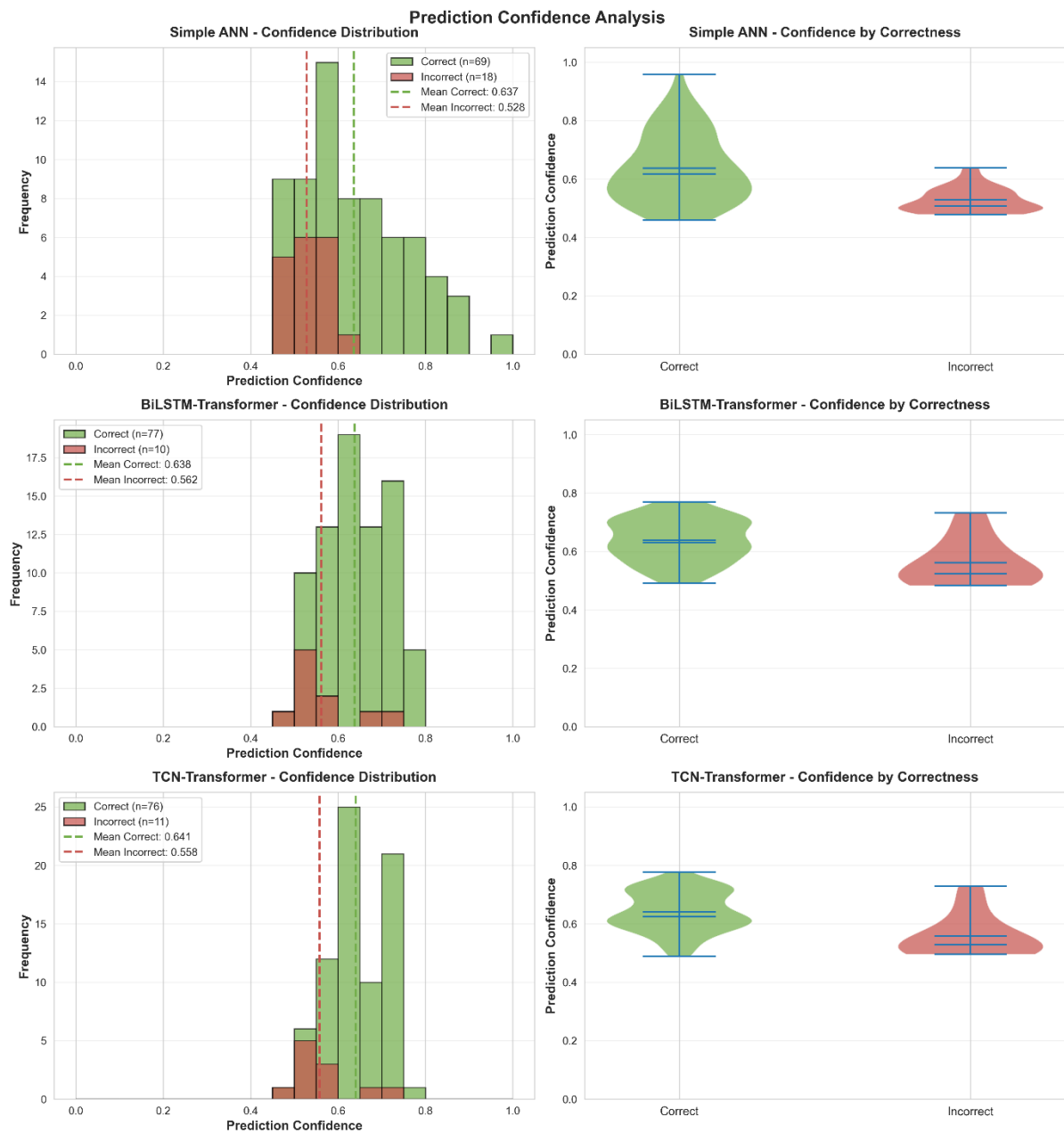


Figure 33: Prediction confidence distributions and violin plots for correct and incorrect predictions across all three imputed models, showing mean confidence values for each group.

The prediction confidence figure presents confidence distributions and violin plots split by correct and incorrect predictions for each model. Across all three models, correctly classified samples had higher mean confidence than incorrectly classified samples. The Simple ANN recorded a mean confidence of 0.637 for correct predictions and 0.528 for incorrect predictions, based on 69 correct and 18 incorrect classifications. The BiLSTM-Transformer recorded mean confidence of 0.638 for correct predictions and 0.562 for incorrect predictions, based on 77 correct and 10 incorrect classifications. The TCN-Transformer recorded mean confidence of 0.641 for correct and 0.558 for incorrect predictions, based on 76 correct and 11 incorrect classifications.

The confidence distributions show that all three models operated in a relatively narrow band, with the majority of predictions between 0.5 and 0.75. None of the models produced a large proportion of confidence predictions above 0.8, and incorrect predictions were not consistently associated with very low confidence scores, with several incorrect predictions made at confidence levels comparable to correct ones. The violin plots illustrated that the spread of confidence was broader for correct predictions than for incorrect ones across all three models.

Calibration

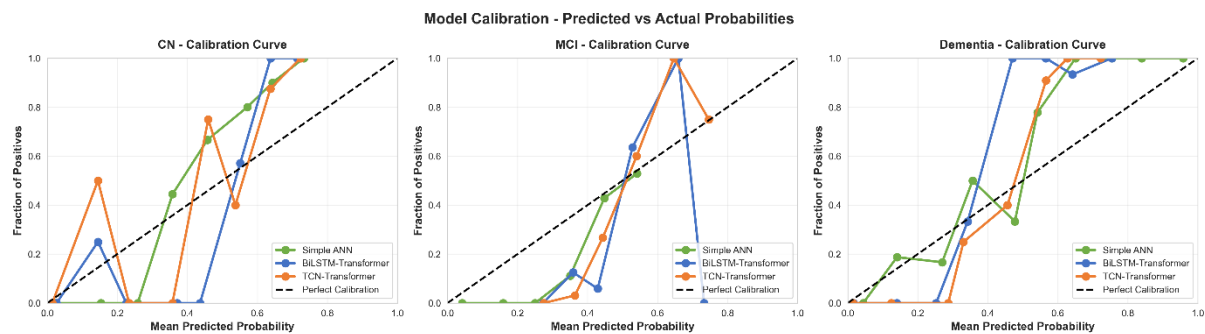


Figure 34: Calibration curves for CN, MCI, and Dementia classes across all three imputed models, with perfect calibration shown as a dashed diagonal.

The calibration curve figure presents predicted versus actual probabilities for each class across all three models, with perfect calibration shown as a dashed diagonal. All three models had notable deviation from perfect calibration across all classes. Rather than tracking the diagonal smoothly, the curves showed erratic behaviour with sharp spikes and drops in the fraction of positives, especially in the low-to-mid predicted probability range. This pattern was most visible for the MCI class, where all three models showed large oscillation around the diagonal, with the BiLSTM-Transformer and TCN-Transformer spiking to a fraction of positives of 1.0 before dropping sharply back toward zero. The Dementia and CN curves showed similar instability, with curves crossing the diagonal multiple times rather than sitting consistently above or below it.

Effect of MICE Imputation

Model Performance Comparison: MICE Imputed vs Unimputed

Metric	BiLSTM-T (Imputed)	BiLSTM-T (Unimputed)	TCN-T (Imputed)	TCN-T (Unimputed)	ANN (Imputed)	ANN (Unimputed)
Test Accuracy	88.51%	82.76%	87.36%	85.06%	79.31%	79.31%
Best Val Accuracy	87.36%	86.21%	83.91%	81.61%	79.31%	78.16%
Best Val Loss	0.1385	0.1365	0.1372	0.1531	0.1911	0.1883
Epochs Trained	30	30	32	32	43	50
MICE Accuracy Gain	+5.75pp	—	+2.30pp	—	+0.00pp	—

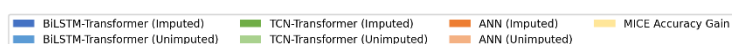


Figure 35: Performance comparison between MICE imputed and unimputed models across all three architectures, including test accuracy, best validation accuracy, best validation loss, epochs trained, and MICE accuracy gain.

The MICE imputed versus unimputed comparison table summarises the effect of imputation on test accuracy across all three models. The BiLSTM-Transformer showed the largest benefit from MICE imputation, with test accuracy improving from 82.76% without imputation to 88.51% with imputation, a gain of 5.75 percentage points. The TCN-Transformer showed a moderate improvement, from 85.06% to 87.36%, a gain of 2.30 percentage points. The Simple ANN showed no change, recording 79.31% in both conditions. The absence of any accuracy gain for the Simple ANN was consistent with its cross-sectional design, which processes only the baseline visit per subject rather than the full longitudinal sequence. As a result, the model was less exposed to the patterns of missing data across visits that MICE imputation was designed to address, unlike the temporal models which relied on complete longitudinal sequences for sequence modelling.

LIME Analysis

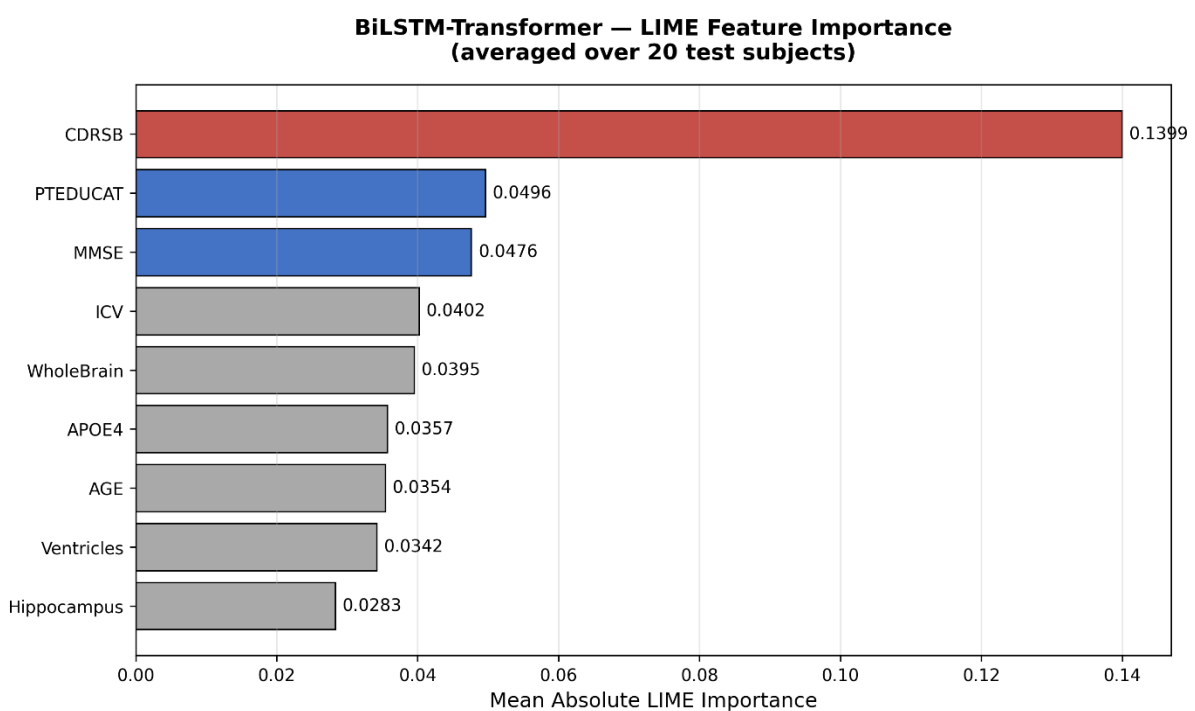


Figure 36: Mean absolute LIME feature importance scores for the BiLSTM-Transformer, averaged across 20 test subjects. CDRSB ranked first with a score of 0.1399, substantially exceeding all other features

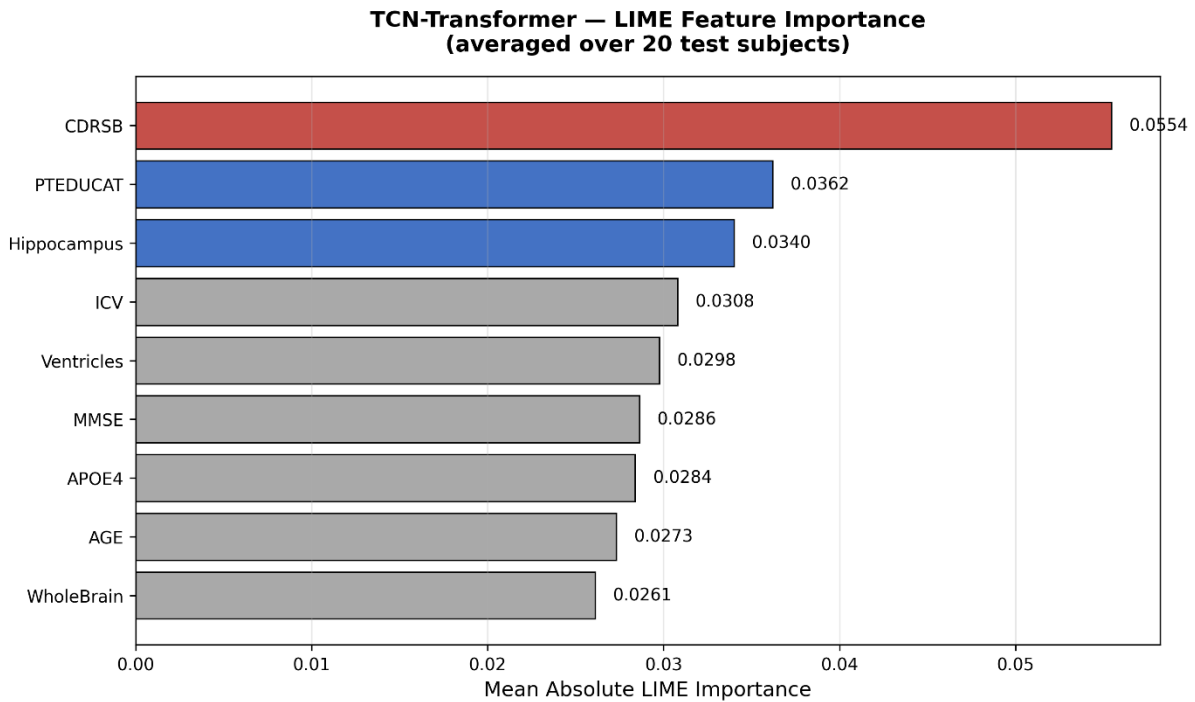


Figure 37: Mean absolute LIME feature importance scores for the TCN-Transformer, averaged across 20 test subjects. CDRSB ranked first (0.0554) with Hippocampus volume ranking third, displacing MMSE from the top three compared to the other models.

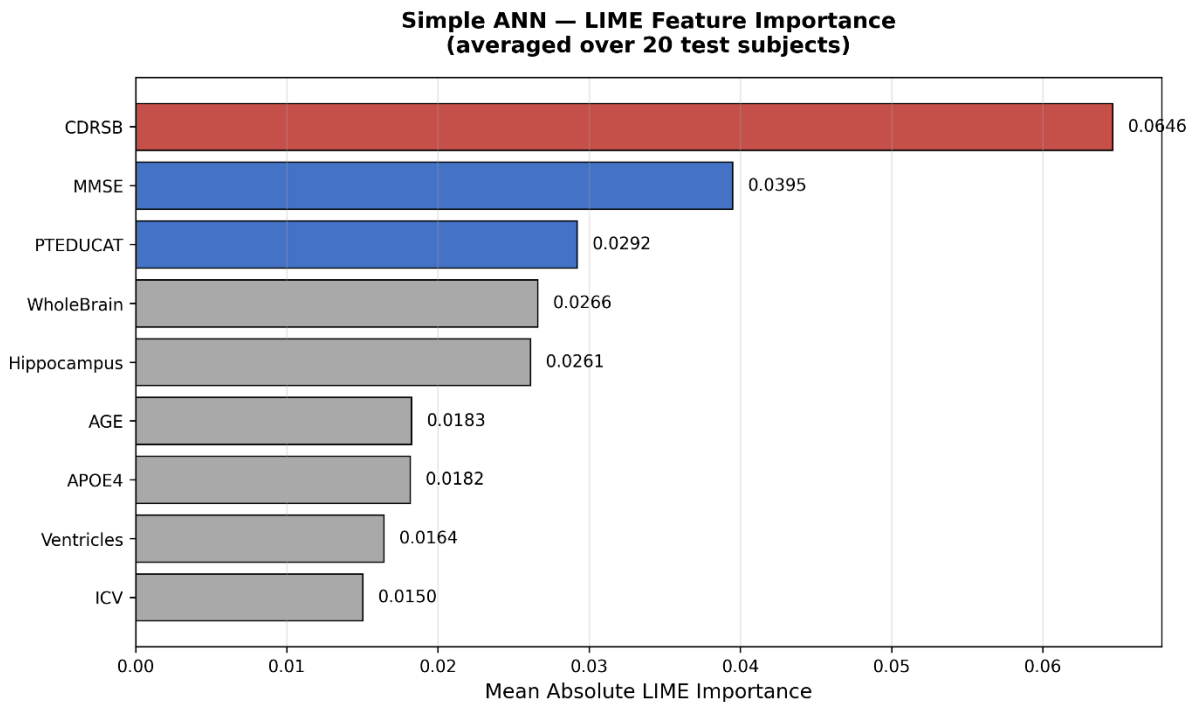


Figure 38: Mean absolute LIME feature importance scores for the Simple ANN, averaged across 20 test subjects. CDRSB ranked first (0.0646) followed by MMSE (0.0395), with the remaining features more evenly distributed.

The LIME analysis was applied to all three trained models following evaluation, with feature importance scores averaged across 20 test subjects per model. The full ranked importance scores for each model are presented in Figures 36, 37, and 38.

CDRSB was the highest ranked feature across all three models, recording mean absolute importance scores of 0.1399 for the BiLSTM-Transformer, 0.0554 for the TCN-Transformer, and 0.0646 for the Simple ANN. For the Simple ANN, MMSE ranked second (0.0395) followed by

PTEDUCAT (0.0292), Hippocampus (0.0261), and WholeBrain (0.0266). For the BiLSTM-Transformer, PTEDUCAT ranked second (0.0496) followed by MMSE (0.0476), ICV (0.0402), and WholeBrain (0.0395). For the TCN-Transformer, PTEDUCAT ranked second (0.0362) followed by Hippocampus (0.0340), ICV (0.0308), and Ventricles (0.0298).

CDRSB was the dominant feature in all three models by a substantial margin, with its importance score exceeding the second-ranked feature by a factor of approximately 2.8 in the BiLSTM-Transformer, and 1.5 in the TCN-Transformer and 1.6 in the Simple ANN. The remaining features were more evenly distributed across positions two through nine in all three models, with no single feature occupying a consistent second or third position across all architectures.

Discussion and Evaluation of Results

Interpretation of Results

BiLSTM vs. TCN

Regarding the comparison between the two temporal models, neither architecture dominated unconditionally across all evaluation metrics. The BiLSTM-Transformer achieved the higher overall test accuracy at 88.51% compared to 87.36% for the TCN-Transformer and led on macro F1 (0.8734 vs. 0.8548), weighted F1 (0.8866 vs. 0.8752), specificity (94.61% vs. 94.05%), and CN recall (96.30% vs. 88.89%). The TCN-Transformer led on macro-AUC (0.9676 vs. 0.9637), Dementia recall (92.31% vs. 87.18%), Dementia F1 (0.947 vs. 0.919), and test loss (0.1372 vs. 0.1385). These different strengths reflect architectural differences rather than noise. The BiLSTM's advantage on overall accuracy and CN classification stems from its bidirectional processing mechanism by reading the visit sequence in both forward and backward directions simultaneously, which allows later visits to contextualise earlier measurements and vice versa, a property identified by Jomeiri et al. (2024) as particularly valuable for retrospective longitudinal classification where the full visit sequence is available at prediction time. By integrating information from both directions simultaneously, stable longitudinal profiles characteristic of CN subjects can be identified with greater precision than a unidirectional or static approach would permit. The TCN's stronger performance on Dementia classification and AUC reflects the ability for dilated causal convolutions to extract temporal features at multiple scales in parallel (Bai et al., 2018), capturing the sustained and progressive pattern of volumetric decline that characterises established dementia across the visit sequence without information needing to propagate sequentially through recurrent gates. Furthermore, the TCN's lead in Dementia recall is consistent with findings by Gopali et al. (2021), who demonstrated that TCN-based models often slightly outperform LSTMs in terms of anomaly detection and identifying critical clinical state transitions in time-series data. The choice between architectures in a clinical context would therefore depend on the specific objective. The BiLSTM-Transformer is preferable where overall accuracy and CN specificity are the priority, while the TCN-Transformer's stronger Dementia recall and AUC profile makes it more appropriate in a screening context where missing an established dementia case carries the greater clinical cost.

Temporal models vs. ANN

The comparison between the temporal models and the Simple ANN baseline provides clear evidence that longitudinal sequence modelling adds meaningful value over cross-sectional deep learning for this task. The BiLSTM-Transformer and TCN-Transformer outperformed the ANN by 9.20% and 8.05% respectively on overall accuracy, but the most clinically significant evidence of this advantage lies in MCI classification. The Simple ANN achieved an MCI recall of 66.67% and an F1-score of 0.609, compared to 80.95% and 0.773 for the BiLSTM-Transformer and 76.19% and 0.744 for the TCN-Transformer. This pattern is consistent with the findings of Bhagwat et al. (2018), who demonstrated that cross-sectional ANN-based approaches consistently failed to identify the slow-declining trajectory class in ADNI longitudinal data, and with Cuingnet et al. (2011), who found that no static classifier performed significantly above chance when predicting MCI-to-AD conversion. The temporal models in this study improve on the ANN baseline most at the diagnostic stage, which is most prone to inaccuracy, confirming that longitudinal sequence modelling captures progressive neuroanatomical change that a single-visit baseline cannot represent. These results compare favourably with Lee et al. (2019), who reported multimodal

longitudinal integration raised MCI conversion prediction accuracy from 75% to 81%, noting that the present study addresses a harder three-class problem.

MICE imputation effect

The effect of MICE imputation further reinforces this interpretation. The BiLSTM-Transformer gained 5.75% from imputation, the TCN-Transformer gained 2.30%, and the Simple ANN gained nothing. The absence of any accuracy gain for the ANN is directly explained by its cross-sectional design, processing only the baseline visits per subject, it is not exposed to the patterns of longitudinal missingness that MICE addresses, and the completeness of a single visit's clinical features was already sufficient for its inference. The temporal models, by contrast, rely on complete longitudinal sequences across multiple visits, and MICE imputation's ability to recover missing MRI-derived features using inter-feature correlations directly strengthens the quality of those sequences. The different imputation benefit between the two temporal models, with the BiLSTM gaining more than twice the accuracy improvement of the TCN, likely reflects its sensitivity to sequence integrity, because it integrates context from both past and future visits simultaneously, gaps at any single timepoint are particularly disruptive to its dynamics (Moghaddami et al., 2025). Consequently, restoring these "broken links" through imputation provides a more substantial performance recovery for the BiLSTM than for the TCN, which utilises parallel dilated convolutions that may be more resilient to sequential gaps. By leveraging MICE to maintain high diagnostic accuracy despite significant missingness, the present study demonstrates a robust approach to the challenging three-class classification task, which is widely recognised as more realistic but significantly more difficult than binary conversion prediction (Aghili et al., 2022).

LIME and Clinical Relevance

CDRSB's dominance across all three models is consistent with Bhagwat et al. (2018), who identified it as the most discriminative feature in ADNI-based classification across multiple independent feature selection methods, and its position as the primary driver of predictions across three architecturally distinct classifiers strengthens confidence that this reflects a genuine clinical signal rather than a model-specific artefact. The BiLSTM-Transformer's CDRSB importance score of 0.1399 was notably higher than both the TCN-Transformer (0.0554) and Simple ANN (0.0646), suggesting that bidirectional sequence processing may increase sensitivity to the longitudinal trajectory of dementia severity as captured by CDRSB more than parallel convolutional or static encoding does, consistent with the BiLSTM's stronger overall accuracy and CN recall.

MMSE ranked within the top three for both the Simple ANN and BiLSTM-Transformer, consistent with its established role as a complementary measure of general cognitive function and a widely used staging indicator across the AD classification literature (Porsteinsson et al., 2021). Hippocampus volume ranked fifth for the Simple ANN and ninth for the BiLSTM-Transformer but rose to third for the TCN-Transformer, where it displaced MMSE from the top three entirely. Hippocampal atrophy is one of the earliest and most reliable neuroanatomical markers of AD progression, with volumetric decline strongly predictive of MCI-to-AD conversion (Jack et al., 2018; Gray et al., 2013). Its prominence in the TCN-Transformer's ranking may reflect the capacity of dilated causal convolutions to capture associations between volumetric atrophy patterns and disease class more directly than recurrent or static approaches. APOE4, the only genetic risk indicator in the feature set, ranked mid-table across all three models, at seventh for the Simple ANN, sixth for the BiLSTM-Transformer, and seventh for the TCN-Transformer. While APOE4 status

is the strongest known genetic risk factor for late-onset AD (Gray et al., 2013), its consistently lower ranking relative to cognitive and neuroimaging features suggests the models weight direct functional and anatomical decline markers more heavily than genetic risk stratification alone.

The consistency of the overall LIME feature importance hierarchy across all three architecturally distinct models is itself a substantive finding. Despite fundamental differences in temporal modelling mechanism, all three have CDRSB as the primary driver and place established cognitive and neuroimaging markers within the top five features, suggesting the models make use of clinically meaningful signals rather than statistical regularities specific to any one architecture. It is important, however, to interpret these results in light of a methodological constraint, LIME operated on the clinical stream only, with baseline visit features replicated across all visit slots during perturbation. This means the analysis captures model sensitivity to variation in baseline clinical values rather than to longitudinal change across visits, which is the primary theoretical advantage of the temporal models. As Amann et al. (2022) and Ribeiro et al. (2016) caution, post-hoc local approximations may not be fully faithful to complex non-linear decision boundaries, and these explanations should therefore be interpreted as indicative rather than definitive.

Calibration and Clinical Deployment

All three models showed notable deviation from ideal calibration across all classes, with erratic oscillation around the perfect calibration diagonal rather than smooth tracking. This was most pronounced for the MCI class, where the BiLSTM-Transformer and TCN-Transformer both spiked to a fraction of positives of 1.0 before dropping sharply back toward zero, and all three models showed large oscillation rather than the gradual curve expected of a well-calibrated classifier. The Dementia and CN curves showed similar instability, crossing the diagonal multiple times across the predicted probability range.

These calibration limitations have direct implications for clinical deployment. A well-calibrated model is one where a predicted probability of, for example, 0.8 for Dementia reflects a true empirical frequency of 80% Dementia cases at that confidence level. Without this property, raw model output probabilities cannot be directly communicated to clinicians as reliable risk estimates or used to support threshold-based clinical decision making. The moderate confidence band seen across all three models further suggests that none of the models express strong probabilistic certainty even when correct, limiting their utility as standalone decision support tools in their current form. The use of Focal Loss as the training objective likely contributes to this miscalibration, as its mechanism of down-weighting well-separated examples compresses the predicted probability distribution toward the decision boundary, a known side effect of focal-type losses that has been documented in the literature (Lin et al., 2018).

Post-hoc calibration techniques should therefore be applied before any clinical translation is considered. Temperature scaling is a lightweight approach that divides logits by a learned scalar parameter before the SoftMax operation and has been shown to substantially improve calibration without degrading discriminative performance (Guo et al., 2017). Platt scaling offers an alternative that fits a logistic regression layer over model outputs (Platt, 1999) and may be better suited to the smaller test set sizes typical of clinical research. Either approach would represent a natural and low-cost enhancement to the models developed in this study prior to any deployment context.

Confidence in Findings and Generalisability

The findings of this study are considered reliable within the scope of the experimental design. The use of a shared 3D CNN encoder, identical training configuration, and fixed random seed across all three architectures ensures that the majority of performance differences observed between models are attributable to temporal modelling mechanism rather than confounding variation in feature extraction or training dynamics. One exception is the TCN's handling of variable-length sequences, where masking is applied only at the aggregation stage rather than during computation, meaning padded timesteps may still influence intermediate representations. This relative difference to the BiLSTM's packed sequence processing is acknowledged as a minor confound in the comparative results and is addressed further in the Limitations section. This controlled design provides a strong basis for the comparative claims made in this study, and the results are competitive with comparable longitudinal multimodal studies.

Generalisation beyond the ADNI cohort is, however, uncertain. ADNI operates under a highly standardised acquisition protocol, and the cohort is predominantly composed of participants from North American clinical research settings, limiting both protocol and demographic generalisability. Performance on data acquired under real-world conditions, different MRI scanners, varying field strengths, or diverse clinical populations, remains unknown. External validation on an independent dataset such as OASIS-3 is the natural next step to assess whether the performance advantages of longitudinal modelling and MICE imputation observed here are reproducible across diverse clinical settings.

Limitations of the Study

Several limitations of this study warrant acknowledgement. First, the dataset comprised 580 subjects from a single research cohort with 87 test subjects, which, while sufficient to support the controlled architectural comparison conducted here, limits both statistical power and the potential for generalisation. Longitudinal clinical datasets of this scale are a practical constraint of the domain rather than a study-specific oversight, but future work should seek to replicate these findings on larger and more diverse cohorts.

Second, the TCN architecture contains a masking asymmetry that represents a methodological limitation specific to this study. Masking is applied at the aggregation stage to exclude padded timesteps from mean pooling, but padded visits may still influence intermediate convolutional representations during the forward pass. Approximately 26.2% of padding was present across sequences in this dataset, meaning this asymmetry is not negligible. This arises from a fundamental property of convolutional architectures as unlike recurrent models, TCNs apply filters across the entire input sequence simultaneously, meaning there is no natural point at which padding can be excluded mid-computation without restructuring the convolutional operations themselves. Addressing this properly would require either causal masking applied at every convolutional layer or repadding the sequence after each layer, both of which add considerable architectural complexity and were beyond the scope of this study. The BiLSTM-Transformer avoids this issue entirely through packed sequence processing, which excludes padded positions from the LSTM computation at the point of input, representing a more rigorous approach to variable-length sequence handling that is more naturally suited to recurrent than convolutional temporal architectures.

Third, the LIME analysis was constrained to the clinical tabular stream, using baseline visit features replicated across all visit slots. While this provides interpretable feature importance

scores that align with clinical knowledge, it does not capture the contribution of the MRI stream to model predictions, nor does it reflect the longitudinal sensitivity that distinguishes the temporal models from the Simple ANN baseline. The explanations produced therefore represent a partial view of model behaviour rather than a complete characterisation.

Fourth, the small test set of 87 subjects, comprising only 21 MCI samples, means that per-class metrics for MCI carry greater statistical uncertainty than those for CN and Dementia. While Focal Loss with class weighting was applied during training to mitigate the effect of class imbalance on model learning, these mechanisms address the training distribution rather than the statistical reliability of evaluation metrics. With only 21 MCI test samples, a single misclassification shifts MCI recall by approximately 4.8 percentage points, meaning small fluctuations in model behaviour produce disproportionately large changes in reported metrics. The stepped appearance of the precision-recall curves is a direct consequence of this limited sample size rather than a reflection of model behaviour at scale.

Evaluation of the Project Process

The TCN-Transformer was implemented and validated first, with the BiLSTM-Transformer being completed afterwards. The addition of a Simple ANN baseline was not included in the original TOR and represented a significant scope addition, motivated by the need for a non-temporal deep learning baseline against which the temporal models could be directly compared.

In retrospect, architecture diagrams generated iteratively alongside implementation rather than retrospectively would have reduced writing-phase pressure. K-fold cross-validation would have provided more statistically robust performance estimates than the single 70/15/15 split used, though this was constrained by the GPU memory limitations when loading full 3D MRI volumes per batch.

Legal, ethical, professional

All legal, ethical, and professional obligations were met throughout the project. The ADNI Data Use Agreement was adhered to in full, with data used exclusively for non-commercial research purposes and not redistributed at any point. Northumbria University ethics approval was obtained prior to data collection, and all identifiable data was stored exclusively on a secured personal device and was not uploaded to any external platform or version control system. No individual subjects were identified or evaluated at any point, with all analysis conducted at the aggregate level. The models produced in this study represent research tools only and carry no intent of clinical deployment without further independent validation.

Conclusion and Recommendations

Research Summary

This study developed and evaluated three deep learning architectures for three-class Alzheimer's Disease staging using longitudinal MRI and clinical data from the ADNI dataset. A BiLSTM-Transformer, TCN-Transformer, and Simple ANN baseline were trained under identical conditions, with MICE imputation applied to address missing data.

The BiLSTM-Transformer achieved the highest test accuracy (88.51%), outperforming the TCN-Transformer (87.36%) and ANN (79.31%). No temporal model consistently outperformed the other, as the BiLSTM led on accuracy, F1, and CN classification, whereas the TCN showed better AUC, Dementia recall, and lower test loss. Both temporal models significantly outperformed the ANN, especially in MCI classification, where recall improved from 66.67% (ANN) to 80.95% (BiLSTM).

MICE imputation improved performance for both temporal models (BiLSTM +5.75%, TCN +2.30%) but had no benefit to the ANN. LIME analysis confirmed reliance on clinically relevant features, with CDRSB consistently dominant.

Contribution

This study makes two primary contributions to AD classification literature. First, it provides the first direct controlled comparison of BiLSTM-Transformer and TCN-Transformer hybrid architectures on longitudinal ADNI data within a single experiment, addressing a gap identified in the literature review. Second, the two-level experimental design, comparing both temporal models against a cross-sectional ANN baseline under identical conditions, provides direct empirical evidence that longitudinal sequence modelling adds meaningful value over static deep learning for three-class AD staging, particularly at the clinically challenging MCI stage. Together these contributions provide both a performance benchmark and an architectural comparison baseline for future hybrid temporal approaches applied to longitudinal AD classification.

Recommendations

Several directions for future work are identified. External validation on OASIS-3 is the most immediate priority to assess whether the performance advantages observed here generalise beyond ADNI. K-fold cross-validation should replace the single holdout split to produce more statistically robust performance estimates. Post-hoc calibration via temperature scaling should be applied before any clinical translation is considered. Extending the task to a four-class CN/sMCI/pMCI/Dementia formulation would better capture MCI heterogeneity and address the most clinically significant classification challenge identified in this study. Finally, applying LIME to the MRI stream through gradient-based attribution methods would provide a more complete characterisation of model decision-making than the clinical-stream-only analysis conducted here.

References

- Aghili, M., Tabarestani, S. and Adjouadi, M. (2022). Addressing the missing data challenge in multi-modal datasets for the diagnosis of Alzheimer's disease. *Journal of Neuroscience Methods*, [online] 375, p.109582. doi:<https://doi.org/10.1016/j.jneumeth.2022.109582>.
- Alatrany, A.S., Khan, W., Hussain, A., Kolivand, H. and Al-Jumeily, D. (2024). An explainable machine learning approach for Alzheimer's disease classification. *Scientific Reports*, [online] 14(1). doi:<https://doi.org/10.1038/s41598-024-51985-w>.
- Amann, J., Vetter, D., Blomberg, S.N., Christensen, H.C., Coffee, M., Gerke, S., Gilbert, T.K., Hagendorff, T., Holm, S., Livne, M., Spezzatti, A., Strümke, I., Zicari, R.V. and Madai, V.I. (2022). To explain or not to explain?—Artificial intelligence explainability in clinical decision support systems. *PLOS Digital Health*, [online] 1(2), p.e0000016. doi:<https://doi.org/10.1371/journal.pdig.0000016>.
- Amatriain, X., Sankar, A., Bing, J., Bodigutla, P.K., J. Hazen, T. and Kazi, M. (2023). Transformer models: an introduction and catalog. *arXiv:2302.07730 [cs]*. [online] Available at: <https://arxiv.org/abs/2302.07730> [Accessed 16 Mar. 2026].
- Anderson, N.D. (2019). State of the science on mild cognitive impairment (MCI). *CNS Spectrums*, [online] 24(1), pp.78–87. doi:<https://doi.org/10.1017/s1092852918001347>.
- Aracri, F., Bianco, M.G., Quattrone, A. and Sarica, A. (2025). Bridging the Gap: Missing Data Imputation Methods and Their Effect on Dementia Classification Performance. *Brain Sciences*, [online] 15(6), p.639. doi:<https://doi.org/10.3390/brainsci15060639>.
- Atri, A. (2019). The Alzheimer's Disease Clinical Spectrum. *Medical Clinics of North America*, [online] 103(2), pp.263–293. doi:<https://doi.org/10.1016/j.mcna.2018.10.009>
- Austin, P.C., White, I.R., Lee, D.S. and van Buuren, S. (2021). Missing Data in Clinical research: a Tutorial on Multiple Imputation. *Canadian Journal of Cardiology*, [online] 37(9). doi:<https://doi.org/10.1016/j.cjca.2020.11.010>.
- Bahrour, S. and Jeon, G. (2026). Longitudinal Alzheimer's Disease Progression Modelling via Hybrid Vision Transformers and Recurrent Neural Networks With Cross-Modal Feature Fusion. *Expert Systems*, [online] 43(3). doi:<https://doi.org/10.1111/exsy.70191>.
- Bai, S., Kolter, J.Z. and Koltun, V. (2018). *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. [online] arXiv.org. Available at: <https://arxiv.org/abs/1803.01271> [Accessed 16 Mar. 2026].
- Bhagwat, N., Viviano, J.D., Voineskos, A.N. and Chakravarty, M.M. (2018). Modeling and prediction of clinical symptom trajectories in Alzheimer's disease using longitudinal data. *PLOS Computational Biology*, [online] 14(9), p.e1006376. doi:<https://doi.org/10.1371/journal.pcbi.1006376>.
- Chen, Q., Hong, Y., Fu, Q. and Bai, H. (2023). *Longformer: Longitudinal Transformer for Alzheimer's Disease Classification with Structural MRIs*. [online] arXiv.org. doi:<https://doi.org/10.48550/arXiv.2302.00901>.
- Cuingnet, R., Gerardin, E., Tessieras, J., Auzias, G., Lehéricy, S., Habert, M.-O., Chupin, M., Benali, H. and Colliot, O. (2011). Automatic classification of patients with Alzheimer's disease from

structural MRI: A comparison of ten methods using the ADNI database. *NeuroImage*, [online] 56(2), pp.766–781. doi:<https://doi.org/10.1016/j.neuroimage.2010.06.013>.

Cummings, J.L., Morstorf, T. and Zhong, K. (2014). Alzheimer's disease drug-development pipeline: few candidates, frequent failures. *Alzheimer's Research & Therapy*, [online] 6(4), p.37. doi:<https://doi.org/10.1186/alzrt269>.

DeTure, M.A. and Dickson, D.W. (2019). The Neuropathological Diagnosis of Alzheimer's Disease. *Molecular Neurodegeneration*, [online] 14(1), pp.1–18. doi:<https://doi.org/10.1186/s13024-019-0333-5>.

Ding, H., Wang, B., Hamel, A.P., Melkonyan, M., Ting, Au, R. and Lin, H. (2023). Prediction of progression from mild cognitive impairment to Alzheimer's disease with longitudinal and multimodal data. *Frontiers in dementia*, [online] 2. doi:<https://doi.org/10.3389/frdem.2023.1271680>.

Dyrba, M., Moritz Hanzig, Slawek Altenstein, Bader, S., Tommaso Ballarini, Brosseron, F., Bürger, K., Cantré, D., Dechent, P., Dobisch, L., Emrah Düzel, Ewers, M., Klaus Fließbach, Glanz, W., Haynes, J., Heneka, M.T., Janowitz, D., Deniz Baris Keles, Ingo Kilimann and Laske, C. (2021). Improving 3D convolutional neural network comprehensibility via interactive visualization of relevance maps: evaluation in Alzheimer's disease. *Alzheimer's Research & Therapy*, [online] 13(1). doi:<https://doi.org/10.1186/s13195-021-00924-2>.

El-Sappagh, S., Saleh, H., Sahal, R., Abuhmed, T., Islam, S.M.R., Ali, F. and Amer, E. (2021). Alzheimer's disease progression detection model based on an early fusion of cost-effective multimodal data. *Future Generation Computer Systems*, [online] 115, pp.680–699. doi:<https://doi.org/10.1016/j.future.2020.10.005>.

Fabietti, M., Mahmud, M., Lotfi, A., Alessandro Leparulo, Fontana, R., Stefano Vassanelli and Fasolato, C. (2023). Early Detection of Alzheimer's Disease from Cortical and Hippocampal Local Field Potentials using an Ensembled Machine Learning Model. [online] pp.1–1. doi:<https://doi.org/10.1109/tnsre.2023.3288835>.

GBD 2019 Dementia Forecasting Collaborators (2022). Estimation of the Global Prevalence of Dementia in 2019 and Forecasted Prevalence in 2050: an Analysis for the Global Burden of Disease Study 2019. *The Lancet Public Health*, [online] 7(2). doi:[https://doi.org/10.1016/s2468-2667\(21\)00249-8](https://doi.org/10.1016/s2468-2667(21)00249-8).

Gers, F.A., Schmidhuber, J. and Cummins, F. (1999). Learning to forget: continual prediction with LSTM. *9th International Conference on Artificial Neural Networks: ICANN '99*. [online] doi:<https://doi.org/10.1049/cp:19991218>.

Gopali, S., Abri, F., Siami-Namini, S. and Namin, A.S. (2021). A Comparative Study of Detecting Anomalies in Time Series Data Using LSTM and TCN Models. *arXiv:2112.09293 [cs]*. [online] Available at: <https://arxiv.org/abs/2112.09293> [Accessed 26 Mar. 2026].

Gray, K.R., Aljabar, P., Heckemann, R.A., Hammers, A. and Rueckert, D. (2013). Random forest-based similarity measures for multi-modal classification of Alzheimer's disease. *NeuroImage*, [online] 65, pp.167–175. doi:<https://doi.org/10.1016/j.neuroimage.2012.09.065>.

Guo, C., Pleiss, G., Sun, Y. and Weinberger, K.Q. (2017). On Calibration of Modern Neural Networks. *arXiv:1706.04599 [cs]*. [online] Available at: <https://arxiv.org/abs/1706.04599> [Accessed 20 Apr. 2026].

Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, [online] 9(8), pp.1735–1780. doi:<https://doi.org/10.1162/neco.1997.9.8.1735>.

Hong, X., Lin, R., Yang, C., Zeng, N., Cai, C., Gou, J. and Yang, J. (2019). Predicting Alzheimer's Disease Using LSTM. *IEEE Access*, [online] 7, pp.80893–80901. doi:<https://doi.org/10.1109/access.2019.2919385>.

Jack, C.R., Bennett, D.A., Blennow, K., Carrillo, M.C., Dunn, B., Haeberlein, S.B., Holtzman, D.M., Jagust, W., Jessen, F., Karlawish, J., Liu, E., Molinuevo, J.L., Montine, T., Phelps, C., Rankin, K.P., Rowe, C.C., Scheltens, P., Siemers, E., Snyder, H.M. and Sperling, R. (2018). NIA-AA research framework: Toward a biological definition of alzheimer's disease. *Alzheimer's & Dementia : the Journal of the Alzheimer's Association*, [online] 14(4), pp.535–562. doi:<https://doi.org/10.1016/j.jalz.2018.02.018>.

Jomeiri, A., Navin, A.H. and Shamsi, M. (2024). Longitudinal MRI analysis using a hybrid DenseNet-BiLSTM method for Alzheimer's disease prediction. *Behavioural brain research*, [online] pp.114900–114900. doi:<https://doi.org/10.1016/j.bbr.2024.114900>.

Kumar, A., Sidhu, J., Goyal, A. and Tsao, J.W. (2022). *Alzheimer Disease*. [online] PubMed. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK499922>.

Lazli, L. (2025). Improved Alzheimer Disease Diagnosis With a Machine Learning Approach and Neuroimaging: Case Study Development. *JMIRx Med*, [online] 6, pp.e60866–e60866. doi:<https://doi.org/10.2196/60866>.

Lee, G., Nho, K., Kang, B., Sohn, K.-A. and Kim, D. (2019). Predicting Alzheimer's disease progression using multi-modal deep learning approach. *Scientific Reports*, [online] 9(1). doi:<https://doi.org/10.1038/s41598-018-37769-z>.

Lei, B., Yang, M., Zhou, F., Hou, W., Zou, W., Li, X., Wang, T., Xiao, X., Wang, S. and Yang, P. (2020). Deep and joint learning of longitudinal data for Alzheimer's disease prediction. *Pattern Recognition*, [online] 102, p.107247. doi:<https://doi.org/10.1016/j.patcog.2020.107247>.

Lin, T.-Y., Goyal, P., Girshick, R., He, K. and Dollár, P. (2018). Focal Loss for Dense Object Detection. *arXiv:1708.02002 [cs]*. [online] Available at: <https://arxiv.org/abs/1708.02002> [Accessed 2 Apr. 2026].

Lo, R.Y. and Jagust, W.J. (2012). Predicting missing biomarker data in a longitudinal study of Alzheimer disease. *Neurology*, [online] 78(18), pp.1376–1382. doi:<https://doi.org/10.1212/wnl.0b013e318253d5b3>.

López-Martos, D., Suárez-Calvet, M., Milà-Alomà, M., Gispert, J.D., Salvadó, G., Brugulat-Serrat, A., Shekari, M., Minguillon, C., Zetterberg, H., Blennow, K., Grau-Rivera, O., Sánchez-Benavides, G., Anastasi, F., Beteta, A., Cacciaglia, R., Mayoral, I.C., Cañas, A., Campo, M. del, Deulofeu, C. and Dominguez, R. (2025). Longitudinal Associations of Multimodal Core 1 Alzheimer Disease Biomarkers With Cognition in Aging and Preclinical Alzheimer Disease. *Neurology*, [online] 105(9). doi:<https://doi.org/10.1212/wnl.0000000000214308>.

Moghaddami, M., Schubring, C. and Siadat, M.-R. (2025). *Transformer Model for Alzheimer's Disease Progression Prediction Using Longitudinal Visit Sequences*. [online] arXiv.org. Available at: <https://arxiv.org/abs/2507.03899> [Accessed 16 Mar. 2026].

Pascanu, R., Mikolov, T. and Bengio, Y. (2013). On the difficulty of training Recurrent Neural Networks. *arXiv:1211.5063 [cs]*. [online] Available at: <https://arxiv.org/abs/1211.5063> [Accessed 1 Apr. 2026].

Platt, J.C. (1999). *Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods*. [online] *Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods*. MIT Press. Available at: https://www.researchgate.net/profile/John-Platt-2/publication/2594015_Probabilistic_Outputs_for_Support_Vector_Machines_and_Comparisons_to_Regularized_Likelihood_Methods/links/004635154cff5262d6000000/Probabilistic-Outputs-for-Support-Vector-Machines-and-Comparisons-to-Regularized-Likelihood-Methods.pdf [Accessed 20 Apr. 2026].

Porsteinsson, A.P., Isaacson, R.S., Knox, S., Sabbagh, M.N. and Rubino, I. (2021). Diagnosis of early Alzheimer's Disease: Clinical practice in 2021. *The Journal of Prevention of Alzheimer's Disease*, [online] 8(3), pp.1–16. doi:<https://doi.org/10.14283/jpad.2021.23>.

Qiu, S., Miller, M.I., Joshi, P.S., Lee, J.C., Xue, C., Ni, Y., Wang, Y., De Anda-Duran, I., Hwang, P.H., Cramer, J.A., Dwyer, B.C., Hao, H., Kaku, M.C., Kedar, S., Lee, P.H., Mian, A.Z., Murman, D.L., O'Shea, S., Paul, A.B. and Saint-Hilaire, M.-H. (2022). Multimodal deep learning for Alzheimer's disease dementia assessment. *Nature Communications*, [online] 13(1), p.3404. doi:<https://doi.org/10.1038/s41467-022-31037-5>.

Rasmussen, J. and Langerman, H. (2019). Alzheimer's Disease – Why We Need Early Diagnosis. *Degenerative Neurological and Neuromuscular Disease*, [online] Volume 9(9), pp.123–130. doi:<https://doi.org/10.2147/dnnd.s228939>.

Ribeiro, M.T., Singh, S. and Guestrin, C. (2016). 'Why Should I Trust You?': Explaining the Predictions of Any Classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining - KDD '16*, [online] pp.1135–1144. doi:<https://doi.org/10.1145/2939672.2939778>.

Rosenbacke, R., Melhus, Å., McKee, M. and Stuckler, D. (2024). How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review. *JMIR AI*, [online] 3, pp.e53207–e53207. doi:<https://doi.org/10.2196/53207>.

Saleh, H., Amer, E., Tamer Abuhmed, Ali, A., Ala Al-Fuqaha and Shaker El-Sappagh (2023). Computer aided progression detection model based on optimized deep LSTM ensemble model and the fusion of multivariate time series data. *Scientific Reports*, [online] 13(1). doi:<https://doi.org/10.1038/s41598-023-42796-6>.

Sarica, A., Cerasa, A. and Quattrone, A. (2017). Random Forest Algorithm for the Classification of Neuroimaging Data in Alzheimer's Disease: A Systematic Review. *Frontiers in Aging Neuroscience*, [online] 9. doi:<https://doi.org/10.3389/fnagi.2017.00329>.

Schuster, M. and Paliwal, K.K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, [online] 45(11), pp.2673–2681. doi:<https://doi.org/10.1109/78.650093>.

Song, M., Jung, H., Lee, S., Kim, D. and Ahn, M. (2021). Diagnostic Classification and Biomarker Identification of Alzheimer's Disease with Random Forest Algorithm. *Brain Sciences*, [online] 11(4), p.453. doi:<https://doi.org/10.3390/brainsci11040453>.

Sperling, R.A., Aisen, P.S., Beckett, L.A., Bennett, D.A., Craft, S., Fagan, A.M., Iwatsubo, T., Jack, C.R., Kaye, J., Montine, T.J., Park, D.C., Reiman, E.M., Rowe, C.C., Siemers, E., Stern, Y., Yaffe, K., Carrillo, M.C., Thies, B., Morrison-Bogorad, M. and Wagster, M.V. (2011). Toward defining the preclinical stages of Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's & Dementia*, [online] 7(3), pp.280–292. doi:<https://doi.org/10.1016/j.jalz.2011.03.003>.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I. and Salakhutdinov, R. (2014). Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research*, [online] 15(56), pp.1929–1958. Available at: <https://jmlr.org/papers/v15/srivastava14a.html> [Accessed 2 Apr. 2026].

Suk, H.-I., Lee, S.-W. and Shen, D. (2014). Hierarchical feature representation and multimodal fusion with deep learning for AD/MCI diagnosis. *NeuroImage*, [online] 101, pp.569–582. doi:<https://doi.org/10.1016/j.neuroimage.2014.06.077>.

Tay, L., Siew Chin Ong, Tien and Thaigarajan Parumasivam (2024). Economic Burden of Alzheimer's Disease: A Systematic Review. *Value in Health Regional Issues*, [online] 40, pp.1–12. doi:<https://doi.org/10.1016/j.vhri.2023.09.008>.

Thapaliya, B., Wu, Z., Sapkota, R., Ray, B., Suresh, P., Ghimire, S., Calhoun, V. and Liu, J. (2024). Graph-based deep learning models in the prediction of early-stage Alzheimers. *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference*, [online] 2024, pp.1–5. doi:<https://doi.org/10.1109/EMBC53108.2024.10782267>.

van Buuren, S. (2018). *Flexible Imputation of Missing Data*. [online] <https://stefvanbuuren.name/fimd/>. Available at: <https://stefvanbuuren.name/fimd/sec-FCS.html> [Accessed 13 Mar. 2026].

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. and Polosukhin, I. (2017). *Attention Is All You Need*. [online] Cornell University. Available at: <https://arxiv.org/abs/1706.03762> [Accessed 16 Mar. 2026].

Venugopalan, J., Tong, L., Hassanzadeh, H.R. and Wang, M.D. (2021). Multimodal deep learning models for early detection of Alzheimer's disease stage. *Scientific Reports*, [online] 11(1), p.3254. doi:<https://doi.org/10.1038/s41598-020-74399-w>.

Vimbi, V., Shaffi, N. and Mahmud, M. (2024). Interpreting artificial intelligence models: a systematic review on the application of LIME and SHAP in Alzheimer's disease detection. *Brain informatics*, [online] 11(1). doi:<https://doi.org/10.1186/s40708-024-00222-1>.

WHO (2021). *World failing to address dementia challenge*. [online] World Health Organization. Available at: <https://www.who.int/news/item/02-09-2021-world-failing-to-address-dementia-challenge> [Accessed 4 Mar. 2026].

World Health Organization (2025). *Dementia*. [online] World Health Organization. Available at: <https://www.who.int/news-room/fact-sheets/detail/dementia> [Accessed 4 Mar. 2026].

Appendix

GitHub Link: <https://github.com/a1101b/KV6013.git>

OneDrive Link: [KV6013-main](#)